THE 2-ADIC ZETA VALUE $\zeta_2(7)$ IS IRRATIONAL

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ABSTRACT. Let $\zeta_2(s) = L_2(s, \omega^{1-s})$ be the 2-adic zeta values of Kubota and Leopoldt. We prove that $\zeta_2(7)$ is irrational. The proof follows the approach of Lai, Sprang and Zudilin for $\zeta_2(5)$: Volkenborn integrals of the third derivative of explicit rational functions R_n give linear forms $S_n = \rho_n + Z_n\zeta_2(7)$ with rational coefficients, and Lai's 2-adic irrationality criterion concludes. The rational function R_n has six nested well-poised numerator blocks and pole orders that ramp up from 1 to 4 towards the middle, and it is integrated at the centred point $t + \frac{1}{2} - \frac{n}{2}$; as a result, no zeta value other than $\zeta_2(7)$ occurs. The 2-adic size of S_n is bounded with the Δ-calculus of Sprang and Lai, the size of the coefficients by a residue-level estimate, and the nonvanishing follows from an auxiliary prime $\ell = \frac{19}{20}n - 1$, after Lai and Sprang. The main new ingredient is a bound for the odd part of the common denominator of ρ_n and Z_n, obtained from a cancellation among the residues in each residue class modulo a prime q (a class lemma together with a global residue identity). Two steps of the proof are computer-assisted: the maximum of an explicit potential, certified by interval and exact rational arithmetic, and the value of the denominator exponent, an exact rational computation over finitely many intervals. The theorem is formalized in Lean 4 with Mathlib, without hypotheses and with only the standard axioms; the one input that is cited rather than formalized is the identification, due to Lai, Sprang and Zudilin, of the Volkenborn-integral definition of $\zeta_2(7)$ used in Lean with the Kubota–Leopoldt value. The mathematics, the proofs and the formalization were produced by AI agents under the direction of the author, and have not been refereed by a human expert.

CONTENTS

1. Introduction 1
2. Preliminaries 4
3. The construction and the linear form 7
4. The 2-adic size (Theorem V) 9
5. The odd denominators (Theorems C, K and A) 10
6. The archimedean size (Theorem G) 17

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7. Nonvanishing (Theorem N) 18
8. Proof of the Main Theorem 19
9. The computer-assisted steps 20
10. Formal verification 21
11. AI use and verification 25
References 26

1. Introduction

1.1. **Background.** For a prime p and an integer $s \geq 2$ let

$$\zeta_p(s) := L_p(s, \omega^{1-s}),$$

where $L_p(s,\chi)$ is the Kubota–Leopoldt p-adic L-function and ω is the Teichmüller character (§2.3). The number $\zeta_p(s) \in \mathbb{Q}_p$ is the p-adic limit of $\zeta(k)$ as $k \to s$ p-adically through negative integers $k \equiv s \pmod{p-1}$, and $\zeta_p(s) = 0$ for even s. The odd values $\zeta_p(3), \zeta_p(5), \ldots$ are p-adic analogues of the odd values of the Riemann zeta function.

In the archimedean case, $\zeta(3)$ is irrational by Apéry's theorem [2], infinitely many of the values $\zeta(2k+1)$ are irrational by the theorem of Rivoal and Ball–Rivoal [28, 3], and at least one of $\zeta(5)$, $\zeta(7)$, $\zeta(9)$, $\zeta(11)$ is irrational by a theorem of Zudilin [33]; the irrationality of $\zeta(5)$ is open. On the 2-adic side, Calegari [5] proved that $\zeta_2(3)$ and $\zeta_3(3)$ are irrational, and Beukers [4] gave another proof. Calegari, Dimitrov and Tang proved that $\zeta_2(5)$ is irrational with their arithmetic holonomy method [6, 7]. Lai, Sprang and Zudilin [17] gave a second proof by an explicit construction in the spirit of Apéry, which also bounds the irrationality measure of $\zeta_2(5)$.

Sprang [31] used Volkenborn integrals of rational functions to construct linear forms in 1 and p-adic zeta values, and proved a p-adic analogue of the Ball–Rivoal theorem. Lai and Sprang [16] showed that many of the numbers $\zeta_p(3), \zeta_p(5), \ldots, \zeta_p(s)$ are irrational (at least a constant times $\sqrt{s/\log s}$ of them, for large odd s). The 2-adic analogue of Zudilin's theorem is due to Lai [14, Theorem 1.1]: for every integer $s \geq 0$, at least one of the numbers $\zeta_2(j)$ with j odd and $s+3 \leq j \leq 3s+5$ is irrational; for $s = 3$ this says that at least one of $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$, $\zeta_2(13)$ is irrational. Lai, Lupu and Sprang [15] proved analogous statements for all primes $p \geq 5$. The author's Lean formalizations [10, 11] contain proofs that at least one of $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$ is irrational and that at least one of $\zeta_2(7)$, $\zeta_2(9)$ is irrational; both were produced by AI agents and are registered in the Palomar registry of Lean-verified results [23, 24].

1.2. **The result.** We call an element of $\mathbb{Q}_2$ *rational* if it lies in the image of $\mathbb{Q} \to \mathbb{Q}_2$.

Main Theorem. *The 2-adic zeta value $\zeta_2(7)$ is irrational.*

The normalization of $\zeta_2(7)$ plays no role: the values of Coleman [9] and Furusho [13] differ from $\zeta_2(7)$ by the non-zero rational factor $(1-2^{-7})^{-1}$ (§2.3). The Main Theorem implies the two statements of [10, 11] quoted above, and the cases $1 \leq s \leq 4$ of Lai's theorem (those whose range contains 7).

An unrefereed draft by C. D. Long [18], posted on GitHub in August 2026, claims that $\zeta_2(s)$ is irrational for every odd s with $3 \leq s \leq 29$, which includes $\zeta_2(7)$, by a different method (a hybrid form of arithmetic holonomy). Our proof

does not use that draft. We were unable to verify it: an internal review, carried out with AI assistance and published with the formalization (`docs/long_review.md`), found that its large-prime estimate (the torsor decomposition and the proposition on "torsor Cartier strictness") does not appear to hold for the Eisenstein–Eichler leaf, and that without it the method's margin for $\zeta_2(7)$ is negative; the draft's other ingredients and all of its printed numerical margins were reproduced. The review is not a refereed report, and the author has not been contacted. No refereed proof of the irrationality of $\zeta_2(7)$ is known to us (§1.6).

1.3. **Outline of the proof.** Fix $n \in 40\mathbb{Z}_{>0}$, let $\eta = (\eta_1, \dots, \eta_6)$ with $40\eta = (5, 7, 7, 9, 11, 13)$, and let $a_k \in \{1, 2, 3, 4\}$ $(0 \le k \le n)$ be the pole orders that equal 1, 2, 3 at distance less than $\frac{n}{20}$, $\frac{n}{10}$, $\frac{3n}{20}$ from the ends $\{0, n\}$ and 4 in the middle (§3.1). Put

$$R_n(t) = (2t+n)\,\frac{\prod_{m=1}^{6}\bigl(t+\frac{1}{2}+\eta_m n\bigr)_{(1-2\eta_m)n}}{\prod_{k=0}^{n}(t+k)^{a_k}}, \qquad S_n = \int_{\mathbb{Z}_2} R_n'''\Bigl(t+\frac{1}{2}-\frac{n}{2}\Bigr)\,\mathrm{d}t, \tag{1.1}$$

where $(x)_L$ is the rising factorial and the integral is the Volkenborn integral. The rational function R_n has degree -3 and satisfies $R_n(-n-t) = -R_n(t)$; its poles have order at most 4. These three properties imply that $S_n = \rho_n + Z_n\zeta_2(7)$ with $\rho_n, Z_n \in \mathbb{Q}$, and that no other zeta value occurs (Lemma 3.2). The proof then consists of four estimates, which are combined by Lai's criterion in the form of Lai and Sprang (§8).

- (V) 2-*adic size* (Theorem V, §4): $v_2(S_n) \ge 6.8n + 7 - 10(\lfloor \log_2 \frac{3n}{4} \rfloor + 1)$, by the Δ-calculus of Sprang and Lai [31, 14].
- (A) *Odd denominators* (Theorems C, K and A, §5): the least odd D_n with $D_n\rho_n, D_nZ_n \in \mathbb{Z}[\frac{1}{2}]$ satisfies $\frac{1}{n}\log D_n \le E^* + o(1)$ with $E^* = 3.8808431\ldots$
- (G) *Archimedean size* (Theorem G, §6): $\log\max(|\rho_n|, |Z_n|) \le U^* n + O(\log n)$ with $U^* = 0.70670223$.
- (N) *Nonvanishing* (Theorem N, §7): if $\ell = \frac{19}{20}n - 1$ is prime, then $v_\ell(\rho_n) = -1$ and $v_\ell(Z_n) \ge 0$.

The criterion needs $E^* + U^* < 6.8\log 2 = 4.7134008$; the margin is

$$E^* + U^* - 6.8\log 2 = -0.1258555\ldots \quad \text{per } n.$$

The construction was found by a numerical search among rational functions of this shape; among the features that make it work, three seem essential. First, the *centred evaluation* at $t + \frac{1}{2} - \frac{n}{2}$: the Volkenborn translation formula then produces correction terms at half-integers h with $|2h| \le n-1$ only, and this is what allows us to show that no prime $q > n$ divides the denominators (Theorem C), without any extra zeros of R_n at half-integers. Secondly, the *pole-order ramps*: the orders $1, 2, 3, 4$ near the ends lower both the growth and the denominators, and orders at most 4 leave only $\zeta_2(5)$ and $\zeta_2(7)$ as candidates among the odd zeta values, of which $\zeta_2(5)$ drops out because R_n has degree -3. Thirdly, the *nested well-poised blocks*: six blocks centred at $\frac{n-1}{2}$, of lengths $\frac{3}{4}n$, $\frac{13}{20}n$ (twice), $\frac{11}{20}n$, $\frac{9}{20}n$, $\frac{7}{20}n$, whose total length $3.4n$ balances the total pole order $3.4n + 4$.

The main new ingredient is the denominator bound (A). Classical arguments bound the q-adic valuation of each residue of R_n separately; this gives the "residue

level" $\max(0, 4 - E_\xi)$, where E_ξ is the balance of zeros and poles in the residue class ξ modulo q, and it leads to the exponent 5.08, far above the break-even value 4.0067. We expand the contribution of each residue class modulo q in a local variable and show that it equals, up to higher-order terms, a residue of R_n times an explicit kernel (the *class lemma*, Lemma 5.3). The sum of these residues over all classes vanishes (Lemma 5.4), because R_n has degree -3. Subtracting it gains up to several powers of q in each class at once. The resulting per-prime bound (Theorem K) is a finite computation on the local models of the classes, which depends only on n/q and k_0/q; a continuum version of it is piecewise constant on explicit intervals (Lemmas 5.11–5.13), and the prime number theorem turns it into the exponent E^* (Theorem A).

1.4. **Computer assistance.** The informal proof is computer-assisted at two places (§9): the maximum U^* of an explicit potential on $[0, 1]$ (Lemma 6.6), certified with interval arithmetic and with exact rational arithmetic, and the value of E^* (Theorem A), an exact rational computation of a step function on 1449 explicit intervals plus a periodic tail. Both were redone by independent implementations. Only the inequality $E^* + U^* < 6.8 \log 2$ matters, and it holds with a margin of 2.7% of the 2-adic gain. In the formal proof all numerical inputs are checked by the Lean kernel (§10).

1.5. **Formal verification.** The Main Theorem is formalized in Lean 4 [21] with Mathlib [19]. The Lean theorem `Zeta2.Z7.zeta2_7_irrational` has no hypotheses and depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The Lean development defines $\zeta_2(7)$ through Volkenborn integrals as $J_6/768$, with the same trusted definitions as the registered formalization [11, 24]; the identification of this number with $L_2(7, \omega^{-6})$ is [17, Lemma 2.8], which is cited and not formalized. The formal proof follows the text with slightly weaker constants (growth rate 0.71, denominator rate 3.99). Section 10 gives the statement, the trusted definitions and the trust base.

1.6. **Novelty and related work.** A literature search by AI agents on 24 and 26 September 2026 (arXiv, including the papers citing [14] and the author pages of Lai, Sprang and Zudilin; OpenAlex; Zenodo; GitHub; public talks and blogs; a general web search for "$\zeta_2(7)$") found no refereed paper and no arXiv preprint that proves the irrationality of $\zeta_2(7)$. The only public claim found is the unrefereed draft [18] mentioned in §1.2, which uses a different method. Google Scholar, MathSciNet, the full text of zbMATH and Lai's thesis were not searched, so we cannot rule out other work.

The methods are those of [31, 14, 16, 17]: Volkenborn integrals of rational functions, the Δ-calculus, Lai's criterion and the auxiliary-prime condition of Lai and Sprang [16, Lemma 2.2]. Non-uniform pole and zero structures and "inserted" rational zeros have archimedean ancestors in the work of Zudilin [34] and of Fischler, Sprang and Zudilin [12], and well-poised symmetry goes back to Ball and Rivoal [3] and Rhin and Viola [27]. We did not find prior p-adic uses of the following ingredients, which we therefore believe to be new: the centred evaluation point; the pole-order ramps together with nested well-poised blocks giving a form in 1 and a single zeta value; the residue-class cancellation for the denominators (Lemmas 5.3–5.5 and Theorem 5.6); and the auxiliary prime $\ell = \frac{19}{20}n - 1$. This is a statement about what we found, not a proof of absence.

1.7. **Organization.** Section 2 collects definitions and published tools. Section 3 gives the construction and the linear form. Sections 4, 5, 6 and 7 prove the four estimates, and Section 8 proves the Main Theorem. Section 9 describes the computer-assisted steps and their certification, Section 10 the formalization, and Section 11 how this work was produced with AI agents and what has been verified, and how.

2. Preliminaries

2.1. **Notation.** Throughout, q and ℓ denote primes. We write v_q for the q-adic valuation on $\mathbb{Q}$ and on $\mathbb{Q}_q$, with $v_q(0) = +\infty$, $|x|_q = q^{-v_q(x)}$, and $\mathbb{Z}_{(q)} = \{x \in \mathbb{Q} : v_q(x) \geq 0\}$. The archimedean absolute value is $|\cdot|$ and log is the natural logarithm.

- $(x)_0 = 1$ and $(x)_k = x(x+1)\cdots(x+k-1)$ (rising factorial); $\binom{x}{k} = x(x-1)\cdots(x-k+1)/k!$.
- $s_2(N)$ is the binary digit sum of $N \geq 0$; Legendre's formula gives $v_2(N!) = N - s_2(N)$, and $v_q(m!) = \sum_{e\geq 1} \lfloor m/q^e \rfloor$.
- $[\varepsilon^\lambda]F$ is the coefficient of ε^λ in a power series or Laurent series $F(\varepsilon)$, and $[y^p]$ likewise.
- $\theta(x) = \sum_{q\leq x} \log q$ is Chebyshev's function.
- For an integer $k \geq 1$, k_- is k with its leading binary digit deleted, so $k - k_- = 2^{\lfloor \log_2 k \rfloor}$.

We use the following elementary facts: Kummer's bound $v_q\binom{N}{K} \leq \lfloor \log_q N \rfloor$; among L consecutive odd integers, at least $\lfloor L/q^e \rfloor$ are divisible by q^e (q odd); and for an odd prime q, $\sum_{u=1}^{q-1} u^{-s} \equiv 0 \pmod q$ unless $(q-1) \mid s$.

2.2. **The Volkenborn integral.** We follow [29, Ch. 5], [14, §2.2] and [17, §2.2]. A function $f\colon \mathbb{Z}_2 \to \mathbb{Q}_2$ is *Volkenborn integrable* [32] if the Riemann sums

$$\frac{1}{2^M} \sum_{x=0}^{2^M-1} f(x) \tag{2.1}$$

converge 2-adically as $M \to \infty$; the limit is $\int_{\mathbb{Z}_2} f(t)\,\mathrm{d}t$. The integral is $\mathbb{Q}_2$-linear but not translation invariant. Strictly differentiable functions are integrable [29, p. 264]; in particular rational functions without poles in $\mathbb{Z}_2$ are.

Lemma 2.1 (Translation; [14, Lemma 2.2], [17, Lemma 2.4])**.** *Let f be strictly differentiable on $\mathbb{Z}_2$ and $d \geq 1$ an integer. Then*

$$\int_{\mathbb{Z}_2} f(t+d)\,\mathrm{d}t = \int_{\mathbb{Z}_2} f(t)\,\mathrm{d}t + \sum_{l=0}^{d-1} f'(l), \qquad \int_{\mathbb{Z}_2} f(t-d)\,\mathrm{d}t = \int_{\mathbb{Z}_2} f(t)\,\mathrm{d}t - \sum_{l=-d}^{-1} f'(l).$$

The second formula follows from the first applied to $t \mapsto f(t-d)$. For $s \geq 1$ put

$$J_s := \int_{\mathbb{Z}_2} \frac{\mathrm{d}t}{(t+\frac{1}{2})^s}; \tag{2.2}$$

the integral exists because $-\frac{1}{2} \notin \mathbb{Z}_2$.

2.3. **2-adic zeta values.** We recall [17, §2.3] (see also [14, §2.3] and [8, Ch. 11]). Let ω be the Teichmüller character and $\langle x \rangle = x/\omega(x)$. For $s \neq 1$ and $x \in \mathbb{Q}_2$ with $|x|_2 \geq 4$, the 2-adic Hurwitz zeta function is $\zeta_2(s,x) = \frac{1}{s-1}\int_{\mathbb{Z}_2} \langle t+x \rangle^{1-s}\,\mathrm{d}t$, and the Kubota–Leopoldt L-function of a Dirichlet character χ of conductor dividing M, $4 \mid M$, is $L_2(s,\chi) = \frac{\langle M \rangle^{1-s}}{M} \sum_{0\leq j<M,\, 2\nmid j} \chi(j)\zeta_2(s, j/M)$.

Definition 2.2 ([17, Definition 2.7])**.** For an integer $s \geq 2$, $\zeta_2(s) := L_2(s, \omega^{1-s})$.

This is the normalization of [14, 16, 17]. The values of Coleman [9] and Furusho [13] differ from it by the Euler factor $(1-2^{-s})^{-1}$ [17, §2.3], a non-zero rational number. By [5, Lemma 2.4], as quoted in [17, §2.3], $\zeta_2(s) = 0$ for even $s \geq 2$.

Lemma 2.3 ([17, Lemma 2.8])**.** *For every integer $s \geq 1$, $J_s = s\,2^{s+1}\zeta_2(s+1)$. In particular $J_s = 0$ for odd s, and $J_6 = 768\,\zeta_2(7)$.*

2.4. **The Δ-calculus.** The following device of Sprang [31] and Lai [14, Def. 2.3, Lemmas 2.4–2.5] (see also [17, Lemmas 2.5–2.6]) bounds the 2-adic size of Volkenborn integrals. For $f\colon \mathbb{Z}_2 \to \mathbb{Q}_2$ put

$$\Delta_0(f) := \inf_{k \geq 1} v_2\Big(\frac{f(k)-f(k_-)}{k-k_-}\Big), \qquad \Delta(f) := \min\{1 + v_2(f(0)), \Delta_0(f)\}.$$

Lemma 2.4. (D1) *If f is strictly differentiable, then $v_2\big(\int_{\mathbb{Z}_2} f\big) \geq \Delta(f) - 1$.*

(D2) *If f, g are $\mathbb{Z}_2$-valued, then $\Delta(fg) \geq \min(\Delta f, \Delta g)$; and $\Delta(f+g) \geq \min(\Delta f, \Delta g)$, $\Delta(Cf) = \Delta(f) + v_2(C)$.*

(D3) *A power series $\sum_j a_j t^j$ with $a_j \in \mathbb{Z}_2$, $a_j \to 0$, has $\Delta \geq 0$.*

(D4) *For $x \in \mathbb{Z}$ and $L \geq 1$, $\Delta\big(\binom{t+x}{L}\big) \geq -\lfloor \log_2 L \rfloor$.*

(D5) *$\big(\frac{\mathrm{d}}{\mathrm{d}t}\big)^{\mu}\binom{t+x}{L} = \sum_{r=\mu}^{L} c_{\mu,r}\binom{t+x}{L-r}$ with $c_{\mu,r} = [z^r](\log(1+z))^{\mu}$ and $v_2(c_{\mu,r}) \geq -\mu\lfloor \log_2 r \rfloor$.*

Proof. (D1)–(D4) are [14, Lemmas 2.4–2.5]; we recall the key identities. (D1): $2^{-M}\sum_{k<2^M} f(k) = f(0) + \sum_{l<M}\sum_{k=2^l}^{2^{l+1}-1}(f(k)-f(k_-))/2^{l+1}$, and each summand has valuation $\geq \Delta_0(f) - 1$. (D2) is the product rule for divided differences. (D4) follows from Vandermonde's identity $\frac{f(k)-f(k_-)}{k-k_-} = \sum_{i=1}^{L}\frac{1}{i}\binom{k-k_--1}{i-1}\binom{k_-+x}{L-i}$ for $f = \binom{t+x}{L}$. (D5): $\frac{\mathrm{d}}{\mathrm{d}t} = \log(1+\nabla)$ on the basis $\binom{t+x}{L}$, where $\nabla\binom{t+x}{L} = \binom{t+x}{L-1}$; the coefficient $c_{\mu,r}$ is a sum of products of μ numbers $\pm 1/r_i$ with $r_i \leq r$, times integers. □

2.5. **Irrationality criteria.**

Lemma 2.5 ([14, Lemma 2.1])**.** *Let $\alpha \in \mathbb{Q}_2$, and for n in an infinite set $I \subseteq \mathbb{N}$ let $a_n, b_n \in \mathbb{Z}$ and $L_n = a_n + b_n\alpha$. If $\max(|a_n|, |b_n|)\cdot|L_n|_2 \to 0$ along I and $L_n \neq 0$ for all $n \in I$, then α is irrational.*

Proof. If $\alpha = u/b$ with $u \in \mathbb{Z}$, $b \geq 1$, then $N_n = bL_n \in \mathbb{Z}\setminus\{0\}$, so $1 \leq |N_n|\cdot|N_n|_2 \leq (b+|u|)\max(|a_n|,|b_n|)\,|L_n|_2$, a contradiction. □

The nonvanishing will come from an auxiliary prime, as in condition (2) of [16, Lemma 2.2]: if $\alpha = u/b$ and a prime $\ell \nmid 2b$ satisfies $v_\ell(a_n) < v_\ell(b_n)$ (with $v_\ell(0) = \infty$), then $v_\ell(bL_n) = v_\ell(a_n) < \infty$, so $L_n \neq 0$. We write this out in §8.

2.6. **A valuation lemma.**

Lemma 2.6. *Let $\Pi(\varepsilon) = \prod_\beta (1+\varepsilon/\beta)^{e_\beta}$ be a finite product with $\beta \in \mathbb{Q}^\times$, $v_q(\beta) \geq 0$ and $e_\beta \in \mathbb{Z}$. Then $v_q([\varepsilon^\lambda]\Pi) \geq -\operatorname{Top}_\lambda$, where*

$$\operatorname{Top}_\lambda := \max\Big\{\sum_\beta j_\beta v_q(\beta) :\ \sum_\beta j_\beta = \lambda,\ j_\beta \geq 0,\ j_\beta \leq e_\beta \text{ whenever } e_\beta \geq 0\Big\}.$$

In particular $\operatorname{Top}_\lambda \leq \lambda \max_\beta v_q(\beta)$; and if every β with $v_q(\beta) > 0$ has $v_q(\beta) = 1$ and $e_\beta \geq 0$, and s is the number of such factors counted with multiplicity e_β, then $\operatorname{Top}_\lambda \leq \min(\lambda, s)$.

Proof. $[\varepsilon^\lambda]\Pi = \sum_{\sum j_\beta=\lambda} \prod_\beta \binom{e_\beta}{j_\beta}\beta^{-j_\beta}$ with $\binom{e}{j} \in \mathbb{Z}$, and $\binom{e}{j} = 0$ for $0 \le e < j$. $\square$

2.7. **Prime number theorem.** We use the prime number theorem $\theta(x) = x+o(x)$, Chebyshev's bound $\theta(x) \le 1.02\,x$ for $x > 0$ (which follows from $\theta(x) < 1.01624\,x$, [30, Theorem 9]), and Dirichlet's theorem for primes $\equiv -1 \pmod{38}$. The $o(x)$ is ineffective as used here.

3. The construction and the linear form

3.1. **The rational function.** Fix $n \in 40\mathbb{Z}_{>0}$.

Numerator blocks. Let $\eta = (\eta_1, \dots, \eta_6) = \frac{1}{40}(5, 7, 7, 9, 11, 13)$ and, for $m = 1, \dots, 6$,

$$C_m := [\eta_m n,\ (1-\eta_m)n - 1] \cap \mathbb{Z}.$$

The block C_m has $L_m = (1-2\eta_m)n$ elements, so the lengths are $\frac{3}{4}n$, $\frac{13}{20}n$, $\frac{13}{20}n$, $\frac{11}{20}n$, $\frac{9}{20}n$, $\frac{7}{20}n$, with total $N := \sum_m L_m = 3.4\,n$. Each C_m is invariant under $c \mapsto n-1-c$, and every block contains $[\frac{13}{40}n, \frac{27}{40}n - 1]$. Write $\mu(c) := \#\{m : c \in C_m\}$.

Pole orders. For $0 \le k \le n$ put $\kappa_k := \min(k, n-k)/n$ and

$$a_k := \begin{cases} 1, & \kappa_k < \frac{1}{20}, \\ 2, & \frac{1}{20} \le \kappa_k < \frac{1}{10}, \\ 3, & \frac{1}{10} \le \kappa_k < \frac{3}{20}, \\ 4, & \kappa_k \ge \frac{3}{20}. \end{cases}$$

Thus $a_k = 1$ on $[0, \frac{n}{20} - 1] \cup [\frac{19}{20}n + 1, n]$, $a_k = 2$ on $[\frac{n}{20}, \frac{n}{10} - 1] \cup [\frac{9}{10}n + 1, \frac{19}{20}n]$, $a_k = 3$ on $[\frac{n}{10}, \frac{3}{20}n - 1] \cup [\frac{17}{20}n + 1, \frac{9}{10}n]$, and $a_k = 4$ on $[\frac{3}{20}n, \frac{17}{20}n]$. The profile is symmetric, $a_{n-k} = a_k$, and $\sum_k a_k = N + 4 = 3.4\,n + 4$.

The form. Put

$$R_n(t) := (2t+n)\,\frac{\prod_{m=1}^{6}\prod_{c\in C_m}(t+\frac{1}{2}+c)}{\prod_{k=0}^{n}(t+k)^{a_k}}, \qquad S_n := \int_{\mathbb{Z}_2} R_n'''\Big(t+\frac{1}{2}-\frac{n}{2}\Big)\,\mathrm{d}t. \tag{3.1}$$

In Pochhammer notation,

$$R_n(t) = \frac{(2t+n)\,\mathrm{Num}(t)}{(t)_{n+1}\,(t+\frac{n}{20})_{\frac{9}{10}n+1}\,(t+\frac{n}{10})_{\frac{4}{5}n+1}\,(t+\frac{3n}{20})_{\frac{7}{10}n+1}},$$

where

$$\mathrm{Num}(t) = \Big(t+\frac{1}{2}+\frac{n}{8}\Big)_{\frac{3}{4}n}\Big(t+\frac{1}{2}+\frac{7n}{40}\Big)^2_{\frac{13}{20}n}\Big(t+\frac{1}{2}+\frac{9n}{40}\Big)_{\frac{11}{20}n} \times \Big(t+\frac{1}{2}+\frac{11n}{40}\Big)_{\frac{9}{20}n}\Big(t+\frac{1}{2}+\frac{13n}{40}\Big)_{\frac{7}{20}n}.$$

The numerator consists of six nested well-poised blocks, all centred at $-\frac{n}{2}$ (in the variable t), and the denominator of four nested Pochhammer symbols, which produce the pole-order ramps.

Lemma 3.1 (Basic properties)**.** (a) *Every zero of the numerator is a half-integer $-\frac{1}{2} - c$, except the zero $-\frac{n}{2}$ of $2t+n$, which lowers the order of the pole at $-\frac{n}{2}$ from 4 to 3. We have the partial-fraction expansion*

$$R_n(t) = \sum_{k=0}^{n}\sum_{i=1}^{a_k}\frac{r_{i,k}}{(t+k)^i}, \qquad r_{i,k} = [\varepsilon^{a_k-i}]G_k(\varepsilon), \quad G_k(\varepsilon) := \varepsilon^{a_k}R_n(-k+\varepsilon),$$

where G_k is holomorphic for $|\varepsilon| < 1$ (and $r_{4,n/2} = 0$).

(b) $\deg R_n = 1 + N - \sum_k a_k = -3$*; hence* $c_1 := \sum_k r_{1,k} = \lim_{t\to\infty} tR_n(t) = 0$.

(c) $R_n(-n-t) = -R_n(t)$. *Consequently* $r_{i,n-k} = (-1)^{i+1} r_{i,k}$, *and* $c_i := \sum_k r_{i,k} = 0$ *for even* i.

(d) $a_k \le 4$, *so* $c_i = 0$ *for* $i \ge 5$. *Hence* c_3 *is the only possibly non-zero* c_i.

Proof. (a) and (d) are read off the definitions, and (b) by counting degrees. For (c), the substitution $t \mapsto -n-t$ maps $2t+n$ to $-(2t+n)$, each factor $t+\frac{1}{2}+c$ to $-(t+\frac{1}{2}+(n-1-c))$, and each $t+k$ to $-(t+n-k)$; since the blocks and the profile are symmetric, $R_n(-n-t) = (-1)^{1+N+\sum a_k} R_n(t)$, and $N + \sum_k a_k = 6.8\,n + 4$ is even. Comparing the principal parts at $-k$ and $-(n-k)$ gives $r_{i,n-k} = (-1)^{i+1} r_{i,k}$; for even i the terms of c_i cancel in pairs, and the term $k = \frac{n}{2}$ vanishes. □

Three independent exact implementations confirm $c_1 = c_2 = c_4 = 0$ and $c_3 \neq 0$ at every n computed.

3.2. **The linear form.** Put $d_k := k - \frac{n}{2}$, let H_k be the set of half-integers strictly between 0 and d_k (so $H_{n/2} = \emptyset$), and

$$X(k,h) := \sum_{i=1}^{a_k} (i)_4\, r_{i,k}\, h^{-i-4} \qquad (h \in H_k). \tag{3.2}$$

Lemma 3.2 (The linear form)**.** *We have*

$$\begin{aligned} S_n &= \rho_n + Z_n\, \zeta_2(7), \qquad Z_n = -46080\, c_3, \\ \rho_n &= \sum_{k=0}^{n} \operatorname{sgn}(d_k) \sum_{h\in H_k} X(k,h) \in \mathbb{Q}. \end{aligned} \tag{3.3}$$

Moreover $\rho_n = -2\sum_{k<n/2}\sum_{h\in H_k} X(k,h)$, *and*

$$X(k,h) = 24\, \operatorname{Res}_{t=-k}\, R_n(t)\,(t_e - t)^{-5}, \qquad t_e := -k + h. \tag{3.4}$$

Proof. The poles of $R_n(t+\frac{1}{2}-\frac{n}{2})$ are half-integers, which are not in $\mathbb{Z}_2$, so S_n exists, and $R_n'''(t) = -\sum_{i,k} (i)_3 r_{i,k} (t+k)^{-i-3}$. Apply Lemma 2.1 to $g(t) = (t+\frac{1}{2})^{-s}$, with $g'(l) = -s(l+\frac{1}{2})^{-s-1}$ and the shift d_k: for $d_k \ge 1$, $\int (t+\frac{1}{2}+d_k)^{-s}\,\mathrm{d}t = J_s - s\sum_{l=0}^{d_k-1}(l+\frac{1}{2})^{-s-1}$, and for $d_k \le -1$ it equals $J_s + s\sum_{l=d_k}^{-1}(l+\frac{1}{2})^{-s-1}$. In both cases the correction is $-\operatorname{sgn}(d_k)\, s \sum_{h\in H_k} h^{-s-1}$. With $s = i+3$ and $(i)_3(i+3) = (i)_4$, summation gives

$$S_n = -\sum_i (i)_3\, c_i\, J_{i+3} + \sum_k \operatorname{sgn}(d_k) \sum_{h\in H_k} \sum_i (i)_4\, r_{i,k}\, h^{-i-4}.$$

By Lemma 2.3, $-(i)_3 c_i J_{i+3} = -(i)_4 2^{i+4} c_i \zeta_2(i+4)$. The terms $i = 1, 2, 4$ vanish by Lemma 3.1 (and $\zeta_2(6) = \zeta_2(8) = 0$ as well), and the term $i = 3$ is $-360\cdot 128\, c_3 \zeta_2(7) = -46080\, c_3 \zeta_2(7)$.

For the symmetric form, write $\sum_{h\in H_k} X(k,h) = \sum_{t\in T_k} P_k''''(t)$, where $P_k(t) = \sum_i r_{i,k}(t+k)^{-i}$ is the principal part at $-k$ and $T_k = H_k - k$ is the set of half-integers strictly between $-k$ and $-\frac{n}{2}$. By Lemma 3.1(c), $P_{n-k}''''(t) = -P_k''''(-n-t)$; the map $t \mapsto -n-t$ sends T_{n-k} onto T_k, and $\operatorname{sgn}(d_{n-k}) = -\operatorname{sgn}(d_k)$. So the terms for k and $n-k$ are equal. Finally, for (3.4) put $u = t+k$, so that $t_e - t = h - u$ and $(t_e - t)^{-5}$ is analytic at $u = 0$. Only the principal part of R_n at $-k$ contributes to the residue, and $24(h-u)^{-5} = \sum_{i\ge 1} (i)_4 h^{-i-4} u^{i-1}$, so the residue is $\sum_i r_{i,k}(i)_4 h^{-i-4}/24$. □

Remark 3.3. Formula (3.3) was implemented by five independent programs, which agree. A zeta-free evaluation of S_n (from Mahler series or from the Taylor coefficients of the product form) agrees with $\rho_n - 60c_3J_6$ modulo $2^{v_2(S_n)+64}$ for $n \le 1600$, where J_6 was computed in three independent ways (Bernoulli numbers, a Mahler series, and a precomputed table), which agree to at least 17000 bits. These checks are not used in the proof.

4. The 2-adic size (Theorem V)

We prove a general bound, of which the case needed here is (4.1). It extends the 2-adic estimates of [14, §4] and [17] to shifted blocks and variable pole orders.

Theorem V. *Let $n \ge 1$, $j \ge 0$, $\delta \in \{0,1\}$ and $m \in \mathbb{Z}$. Let $[\mathrm{lo}_\nu, \mathrm{hi}_\nu]$ $(\nu = 1, \dots, B)$ be integer intervals of lengths $L_\nu \ge 0$, put $x_\nu := \mathrm{hi}_\nu + 1 - m$, and let $a_k \ge 0$ $(0 \le k \le n)$ be integers. Put*

$$R(t) := (2t+n)^\delta \, \frac{\prod_\nu \prod_{c=\mathrm{lo}_\nu}^{\mathrm{hi}_\nu} (t + \frac12 + c)}{\prod_k (t+k)^{a_k}}, \qquad S := \int_{\mathbb{Z}_2} R^{(j)}\Big(t + \frac12 - m\Big)\,\mathrm{d}t,$$

$N := \sum_\nu L_\nu$, $\sigma := \sum_\nu s_2(L_\nu)$ and $\lambda := \lfloor \log_2 \max(1, \max_\nu L_\nu) \rfloor + 1$. Then

$$v_2(S) \ge N + \sum_k a_k + j - \sigma - (j+1)\lambda.$$

For R_n we have $B = 6$, $j = 3$, $m = \frac{n}{2}$, $x_\nu = (\frac12 - \eta_\nu)n \ge 0$, $\max_\nu L_\nu = \frac34 n$ and $\sigma \le 6\lambda$, so

$$v_2(S_n) \ge 6.8\,n + 7 - \sigma_n - 4\lambda_n \ge 6.8\,n + 7 - 10\lambda_n, \qquad \lambda_n = \Big\lfloor \log_2 \frac{3n}{4} \Big\rfloor + 1. \tag{4.1}$$

Proof. Factorization. Since $t + \frac12 - m + \frac12 + c = t + 1 + c - m$ and $t + \frac12 - m + k = \frac12(2t + 2k + 1 - 2m)$,

$$R\Big(t + \frac12 - m\Big) = 2^{\sum_k a_k}\, E(t) \prod_\nu L_\nu! \binom{t + x_\nu}{L_\nu},$$

$$E(t) := (2t + 1 + n - 2m)^\delta \prod_k (2t + 2k + 1 - 2m)^{-a_k}.$$

Every factor $2t + b$ with b odd is a 2-adic unit on $\mathbb{Z}_2$, and its inverse is a power series in t with coefficients in $\mathbb{Z}_2$ tending to 0. Hence $E_\beta := E^{(\beta)}/(\beta!\,2^\beta)$ is $\mathbb{Z}_2$-valued with $\Delta(E_\beta) \ge 0$, by (D2) and (D3).

Terms. By Leibniz' rule and (D5), $R^{(j)}(t + \frac12 - m)$ is a sum of terms $K \cdot \Phi(t)$ with $\Phi = E_\beta \prod_\nu \binom{t + x_\nu}{L_\nu - r_\nu}$, where $\beta + \sum_\nu \mu_\nu = j$, $\mu_\nu \le r_\nu \le L_\nu$, and

$$K = 2^{\sum a_k}\, \frac{j!}{\prod_\nu \mu_\nu!}\, 2^\beta \prod_\nu L_\nu!\, c_{\mu_\nu, r_\nu}.$$

By Legendre's formula and (D5), $v_2(K) \ge \sum_k a_k + \beta + (N - \sigma) - (j - \beta)(\lambda - 1)$. By (D2)–(D4), $\Delta(\Phi) \ge -(\lambda - 1)$, so $v_2(\int \Phi) \ge -\lambda$ by (D1). Hence $v_2(K \int \Phi) \ge N + \sum a_k + j - \sigma - (j+1)\lambda + \beta\lambda$, which proves the theorem. □

Remark 4.1. For $n = 480$, 960 and 2000 the bound (4.1), with the exact σ_n, gives 3214, 6474 and 13534, while the true values of $v_2(S_n)$ are 3237, 6499 and 13560; the slack is 18 to 27 bits for all $n = 40, \dots, 2000$. The Lean proof obtains $v_2(S_n) \ge 6.8n + 4 - \sigma_n - 4(\lambda_n - 1)$, one more than (4.1).

5. The odd denominators (Theorems C, K and A)

5.1. **Set-up.** Let D_n be the least odd positive integer with $D_n\rho_n$, $D_nZ_n \in \mathbb{Z}[\frac{1}{2}]$. For an odd prime q put

$$e_q(n) := v_q(D_n) = \max\bigl(0, -v_q(\rho_n), -v_q(Z_n)\bigr).$$

Since $46080 = 2^{10}\cdot 3^2\cdot 5$, we have $v_q(Z_n) \geq v_q(c_3)$ for all odd q, so $e_q(n) \leq \max(0, -v_q(\rho_n), -v_q(c_3))$.

Fix an odd prime q and put $x := n/q$. A pair (k,h) with $h \in H_k$ is *bad* if $q \mid 2h$ (note that $2h$ is an odd integer) and *good* otherwise. Write $\rho_n = G + B$, where G and B are the sums in (3.3) over the good and the bad pairs.

We begin with the primes above n, where the centred evaluation pays off.

Theorem C. *If $q > n$ is an odd prime, then $v_q(\rho_n) \geq 0$ and $v_q(c_3) \geq 0$; hence $q \nmid D_n$.*

Proof. All pole differences satisfy $|k' - k| \leq n < q$, and the values $h \in H_k$ satisfy $|2h| \leq n - 1 < q$, so these are q-units; the factor $n - 2k$ is a q-unit or, for $k = \frac{n}{2}$, is replaced by 2ε, and there we work with $G_k(\varepsilon)/(2\varepsilon)$. The numerator values $2(c-k)+1$ may be divisible by q. Write $G_k(\varepsilon) = G_k(0)\prod_\beta (1+\varepsilon/\beta)^{e_\beta}$, where β runs over the numerator values $c-k+\frac{1}{2}$ (with $e_\beta \geq 0$) and over the pole differences $k'-k$ (with $e_\beta = -a_{k'}$), with the obvious change at $k = \frac{n}{2}$. Since $G_k(0)$ contains the factor β^{e_β} for every numerator value, $G_k(0)\binom{e_\beta}{j_\beta}\beta^{-j_\beta}$ is q-integral for $j_\beta \leq e_\beta$, and it vanishes for $j_\beta > e_\beta$ (compare Lemma 2.6). Hence every $r_{i,k}$ is q-integral, so is every $X(k,h)$ (h is a q-unit), and so are ρ_n and c_3. □

Numerically, no prime in $(n, 2n+2]$ divides D_n for any $n \leq 40000$ tested.

5.2. **Residue discs and local models.** All zeros and poles of R_n lie in $\frac{1}{2}\mathbb{Z} \subset \mathbb{Z}_q$.

Discs. For $k_0 \in \{0, \dots, q-1\}$ let $\xi = \xi(k_0)$ be the class of $-k_0$, with disc $D_\xi := -k_0 + q\mathbb{Z}_q$ and centre $\hat{\xi} := -k_0$. For a rational function F with poles in $\frac{1}{2}\mathbb{Z}$ put $\oint_{D_\xi} F := \sum_{w\in D_\xi} \operatorname{Res}_{t=w} F$. Every pole $-k$ lies in exactly one disc, and the evaluation point $t_e = -k + h$ of a pair lies in the disc of $-k$ if and only if the pair is bad. Hence $B = \sum_\xi \beta_\xi$, where $\beta_\xi := \sum \operatorname{sgn}(d_k) X(k,h)$ over the bad pairs with $k \equiv k_0 \pmod q$.

Items. Write $R_n(t) = 2\prod_w (t-w)^{e_w}$. The *items* w are the poles $-k$ (with $e_w = -a_k$), the zeros $-\frac{1}{2} - c$ (with $e_w = \mu(c)$), and the point $-\frac{n}{2}$ of the factor $2t+n$ (with $e_w = +1$), which we keep separate from the pole at $-\frac{n}{2}$.

Local coordinate. Put $v := (t - \hat{\xi})/q$. An item $w \in D_\xi$ has the local point $v_w := (w - \hat{\xi})/q \in \frac{1}{2}\mathbb{Z}$. Define

$$RR_\xi(v) := \prod_{w\in D_\xi} (v - v_w)^{e_w}, \qquad E_\xi := \sum_{w\in D_\xi} e_w, \qquad \hat{U}_\xi(t) := 2\prod_{w\notin D_\xi} (t-w)^{e_w}.$$

Then $R_n(t) = q^{E_\xi}\,\hat{U}_\xi(t)\,RR_\xi(v)$ on D_ξ.

The *local model* of ξ is the following data: the poles $v = -j$ for $k = k_0 + jq \leq n$, of order a_k; the zeros $v = -s/2$, s odd, with $c = k_0 + (sq-1)/2 \in [0, n)$, of multiplicity $\mu(c)$; the δ-zero $v = (2k_0 - n)/(2q)$, present when $q \mid n - 2k_0$, which sits on the pole $-\frac{n}{2}$ and lowers its net order to 3; and the bad pairs, that is, a pole $v_k = -j$ with $d_k \neq 0$ and an evaluation point $v_e = v_k + \operatorname{sgn}(d_k)\sigma/2$ with

σ odd and $\sigma q < 2|d_k|$. Equivalently, v_e is a half-integer strictly between v_k and $v_c := (2k_0 - n)/(2q)$.

Lemma 5.1 (Outside factor)**.** *$\hat{U}_\xi(\hat{\xi} + qv) = \sum_{m\geq 0} F_m q^m v^m$ with $F_m \in \mathbb{Z}_q$ and $F_0 \in \mathbb{Z}_q^\times$. The series converges for $|v|_q \leq 1$ and equals the rational function there.*

Proof. For $w \notin D_\xi$, $u_w := \hat{\xi} - w$ is a q-unit, and $(u_w + qv)^e = u_w^e \sum_r \binom{e}{r}(qv/u_w)^r$ with $\binom{e}{r} \in \mathbb{Z}$ for every $e \in \mathbb{Z}$. A finite product of such series has the stated form, and $\hat{U}_\xi$ has no pole on $|v|_q \leq 1$. □

Lemma 5.2 (Sizes and q-integrality)**.** *Every local point (pole, zero, δ-point, evaluation point) lies in $[-x - \frac{1}{2}, \frac{1}{2}]$. Consequently every rational number built below from a local model (the Taylor and Laurent coefficients of RR_ξ, of $(v_e - v)^{-5}$ and of $\prod(1 - v_w y)^{e_w}$, polynomial combinations of these, and the kernel constants of Theorem 5.6) has a denominator composed of 2, of integers $2|v - v'| \leq 2x + 2$, and of odd $\sigma < 2x$. All of these numbers are q-integral once $q > 2x + 2$, in particular once $q^2 > 2n + 3q$.*

Proof. Poles satisfy $v = -j \in [-x, 0]$. For zeros, $c \geq 0$ and $k_0 \leq q - 1$ force $s \geq -1$, and $c < n$ forces $s < 2x + 1/q$. Evaluation points lie between a pole and $v_c \leq \frac{1}{2}$. Expanding $RR_\xi(v)(v_e - v)^{-5}$ around v_k only divides by the differences $v_k - v_w$ and $v_e - v_k$. □

5.3. **The class lemma.** For a class ξ let s be a local variable and $y := 1/s$. Put

$$A_\xi(s) := \sum_{\text{bad pairs } (k,e) \text{ of } \xi} \operatorname{sgn}(d_k)\, \mathrm{PP}_{v_k}\big[RR_\xi(v)(v_e - v)^{-5}\big](s),$$

where PP_{v_k} denotes the principal part at v_k, and

$$r_\xi(y) := \prod_{w \in D_\xi} (1 - v_w y)^{e_w} = y^{E_\xi} RR_\xi(1/y) \in \mathbb{Q}[[y]], \qquad r_\xi(0) = 1.$$

For a *kernel polynomial* $\kappa(y) = c_0 + \sum_{j\geq 1} c_j y^j \in \mathbb{Q}[y]$ put $K(v) := \kappa(1/v)$ and

$$D_{\xi,\kappa}(y) := y^{E_\xi} A_\xi(1/y) - r_\xi(y)\, \kappa(y).$$

Lemma 5.3 (Class lemma)**.** *Let $q \geq 5$ with $q > 2x + 2$, and let the c_j be q-integral. Then*

$$\beta_\xi - 24q^{-5} \oint_{D_\xi} R_n(t)\, K\Big(\frac{t - \hat{\xi}}{q}\Big)\, \mathrm{d}t = 24q^{E_\xi - 4} \sum_{m\geq 0} F_m q^m\, [y^{m+E_\xi+1}] D_{\xi,\kappa}(y). \tag{5.1}$$

Consequently the left-hand side has q-adic valuation at least

$$\mathrm{Lev}_\kappa(\xi) := -5 + \min\{p \geq E_\xi + 1 : [y^p] D_{\xi,\kappa} \neq 0\}$$

($= +\infty$ if there is no such p), and always $\mathrm{Lev}_\kappa(\xi) \geq E_\xi - 4$.

Proof. (i) *The bad part.* By (3.4), a bad pair has $X(k,h) = 24 \operatorname{Res}_{t=-k} R_n(t)(t_e - t)^{-5}$. Substituting $t = \hat{\xi} + qv$, so that $\mathrm{d}t = q\,\mathrm{d}v$ and $t_e - t = q(v_e - v)$, gives $X = 24q^{E-4} \operatorname{Res}_{v=v_k}\big[\hat{U}(\hat{\xi} + qv) RR(v)(v_e - v)^{-5}\big]$. Insert Lemma 5.1; the re-expansion at v_k converges in $\mathbb{Q}_q$ because $|v_k|_q \leq 1$. The residue at v_k only sees the principal part $P_{k,e}$ of $RR(v)(v_e - v)^{-5}$, which is a rational function with its only pole at v_k and vanishing at ∞; hence $\operatorname{Res}_{v_k}[v^m P_{k,e}] = -\operatorname{Res}_\infty[v^m P_{k,e}] = [y^{m+1}]P_{k,e}(1/y)$. Summing with signs, $\beta_\xi = 24q^{E-4} \sum_m F_m q^m [y^{m+E+1}]\big(y^E A_\xi(1/y)\big)$.

(ii) *The subtracted part.* The poles of $R_n(t)K(v)$ in D_ξ are the local poles, together with $v=0$ if some $c_j \neq 0$; these are all the finite poles of the rational function $v^m RR(v)K(v)$. Hence $\oint_{D_\xi} R_n K \,\mathrm{d}t = q^{E+1}\sum_m F_m q^m \cdot [v^{-1}]_\infty(v^m RR\,K)$, where $[v^{-1}]_\infty$ is the coefficient in the expansion at ∞. Since $v^m RR(v)K(v) = \sum_p [y^p](r_\xi\kappa)\, v^{m+E-p}$, this coefficient is $[y^{m+E+1}](r_\xi\kappa)$.

Multiply (ii) by $24q^{-5}$ and subtract. All coefficients are q-integral (Lemma 5.2 and $F_m \in \mathbb{Z}_q$), and the m-th term has valuation at least $E-4+m$. □

Three kinds of kernels are used: $\kappa = 0$, which gives the levels $\mathrm{Lev}_0(\xi)$ (a class without bad pairs has $\beta_\xi = 0$); $\kappa = 1$, which gives $\mathrm{Lev}_1(\xi)$; and $\kappa = 1 + c\,y^j$ with $j \geq 0$, which gives $\mathrm{Lev}_\kappa(\xi)$. In the computations the levels are evaluated in exact rational arithmetic with series truncated at order 12: the coefficients are exact below the truncation index, and if no non-zero coefficient is found there, the truncation index is used, which is still a lower bound.

5.4. **The kernel identity and the far part.**

Lemma 5.4 (Kernel identity)**.** *Let $\Psi(t) := c_0 + \sum_{j=1}^{J} c_j q^j \sum_{i=0}^{q-1}(t+i)^{-j}$. Then $\sum_\xi \oint_{D_\xi} R_n \Psi \,\mathrm{d}t = 0$.*

Proof. $R_n\Psi$ is a rational function whose finite poles are integers, each lying in exactly one disc. So the left-hand side is the sum of all finite residues, which equals $-\operatorname{Res}_\infty(R_n\Psi) = 0$ because $\deg(R_n\Psi) \leq \deg R_n = -3 \leq -2$. □

On D_ξ the summand $i = k_0$ of Ψ equals $c_j q^j (t-\hat{\xi})^{-j} = c_j v^{-j}$. Hence $\Psi = K(v) + \sum_{j\geq 1} c_j q^j g_{j,\xi}(t)$ with $g_{j,\xi}(t) := \sum_{i\neq k_0}(t+i)^{-j}$, which is analytic on D_ξ since $0 < |i-k_0| < q$.

Lemma 5.5 (Far part)**.** *Let $q > 2x+2$, $j \geq 1$, c_j q-integral, and $q \geq j+1-E_\xi$. Then*

$$v_q\Big(24q^{-5}\,c_j q^j \oint_{D_\xi} R_n\, g_{j,\xi}\,\mathrm{d}t\Big) \geq j-4.$$

Proof. We have $g_{j,\xi}(\hat{\xi}+qv) = \sum_{r\geq 0}\binom{-j}{r} q^r v^r P_{j+r}$ with $P_s := \sum_{i\neq k_0}(i-k_0)^{-s}$. The numbers $i-k_0$ ($i \neq k_0$) form a complete system of non-zero residues modulo q, so $P_s \equiv 0 \pmod q$ unless $(q-1) \mid s$. Write $\hat{U}\cdot g = \sum_m H_m q^m v^m$. For $m \leq q-2-j$, every P_{j+r} with $r \leq m$ vanishes modulo q, so $q \mid H_m$; for larger m, $v_q(H_m q^m) \geq m \geq q-1-j$. As in part (ii) of the proof of Lemma 5.3, $\oint_{D_\xi} R_n g = q^{E+1}\sum_m H_m q^m [y^{m+E+1}] r_\xi$, and only $m \geq \max(0,-E-1)$ contribute. Each term has valuation at least $E+1+m+1 \geq 1$ (first range) or $E+1+q-1-j \geq 1$ (second range, by the hypothesis on q). Multiplying by $24q^{j-5}c_j$ gives the claim. □

Theorem 5.6 (Collective bound for the bad part)**.** *Let $q \geq 29$ with $q > 2x+2$. Put*

$$L_a := \min_{\xi\ \textit{exposed}} \mathrm{Lev}_0(\xi), \qquad L_b := \min_\xi \mathrm{Lev}_1(\xi),$$

where a class is exposed *if it has a bad pair. Define L_κ as follows. Put $P_0 := L_b$ and $j := P_0+5$. If $j \geq 0$ and every local type of a class with $E_\xi+1 \leq j$ has the same coefficient $c := [y^j]D_{\xi,1} \neq 0$, set $\kappa := 1+c\,y^j$ and $L_\kappa := \min(\min_\xi \mathrm{Lev}_\kappa(\xi), P_0+1)$; otherwise set $L_\kappa := P_0$. Then*

$$v_q(B) \geq \mathfrak{B}_q := \max(L_a, L_b, L_\kappa).$$

Proof. By Lemma 5.4, $B = \sum_\xi \big[\beta_\xi - 24q^{-5} \oint_{D_\xi} R_n\Psi\big]$ for every admissible Ψ: $\Psi = 0$ gives the bound L_a, $\Psi = 1$ gives L_b, and the κ-kernel gives L_κ (for $j = 0$ it is the constant $\Psi = 1 + c$, with no far part; for $j \geq 1$ it is $c_0 = 1$, $c_j = c$). Each bracket is (5.1), plus, for $j \geq 1$, the far part of Lemma 5.5, which costs $j - 4 = P_0 + 1$. For every type with $E + 1 \leq j$ we have $[y^p]D_{\xi,\kappa} = [y^p]D_{\xi,1} = 0$ for $E + 1 \leq p < j$ (by the definition of P_0) and $[y^j]D_{\xi,\kappa} = c - c = 0$, so $\mathrm{Lev}_\kappa \geq P_0 + 1$. Types with $E + 1 > j$ have $\mathrm{Lev}_\kappa \geq -5 + E + 1 \geq P_0 + 1$ whatever κ is. Lemma 5.5 applies, because $E_\xi \geq -10$ (Lemma 5.9) and $j \leq 4$ (the step only matters when $P_0 < 0$), so $q \geq 29 \geq j + 1 - E_\xi$. □

Remark 5.7. Only $j \geq 0$ is admissible. For $j < 0$ the "kernel" would be a polynomial in v, the degree of $R_n\Psi$ could be ≥ -1, and Lemmas 5.4–5.5 would fail. An early version of the computation overlooked this and produced bounds below the true exponent at 19 primes; the verification step detected it.

5.5. **The good part and c_3.**

Lemma 5.8 (Per-pole bound)**.** *Let $q \geq 5$ with $q > 2x + 2$. Let $-k$ be a pole in the class ξ with net local order a ($a = a_k$, or $a = 3$ at $-\frac{n}{2}$), and write $\tau_k(w) := w^a R R_\xi(v_k + w) = \sum_l \tau_l w^l$ ($\tau_0 \neq 0$). Put $\ell_k(i) := \max\{l \leq a - i : \tau_l \neq 0\}$. Then $v_q(r_{i,k}) \geq E_\xi + a - \ell_k(i)$.*

Proof. Expand around $t_k = -k$ as in Lemma 5.1:

$$R_n(t_k{+}qw) = q^E\, \hat{U}(t_k{+}qw)\, w^{-a} \tau_k(w), \qquad \hat{U}(t_k{+}qw) = \sum_m F'_m q^m w^m, \quad F'_m \in \mathbb{Z}_q.$$

So $r_{i,k} = q^{E+i} \sum_{l+m=a-i} F'_m q^m \tau_l$, and every term has valuation at least $E + a - l$ (τ_l is q-integral by Lemma 5.2). □

Let $\mathfrak{G}_q$ be the minimum of $E + a - \ell(i)$ over the local types realized by the q classes, their poles and $i \leq a$, and let $\mathfrak{C}_q$ be the minimum of $E + a - \ell(3)$ over the poles with $a \geq 3$. For $q \geq 11$ the numbers $(i)_4 \in \{24, 120, 360, 840\}$ are q-units, and so is h for a good pair; hence $v_q(G) \geq \mathfrak{G}_q$ and $v_q(c_3) \geq \mathfrak{C}_q$.

5.6. **Theorem K.**

Theorem K. *Let $q \geq 29$ be a prime with $q^2 > 2n + 3q$. Then*

$$e_q(n) \leq f_n(q) := \max\big(0, -\min(\mathfrak{B}_q, \mathfrak{G}_q), -\mathfrak{C}_q\big).$$

Proof. $q^2 > 2n + 3q$ gives $q > 2x + 3$, so Lemmas 5.2–5.8 and Theorem 5.6 apply. Then $\rho_n = G + B$ with $v_q(B) \geq \mathfrak{B}_q$ and $v_q(G) \geq \mathfrak{G}_q$, and $v_q(c_3) \geq \mathfrak{C}_q$. □

The number $f_n(q)$ is a finite exact computation on the local models. It was implemented twice, independently (once directly, once through residue moments modulo the prime $2^{127} - 1$).

5.7. **Crude bounds and the small primes.**

Lemma 5.9 (Crude level bounds and exposure)**.** *For every n, every prime $q \geq 5$ and every class:*

1. *$E_\xi \geq -10$. Hence, if $q \geq 29$ and $q^2 > 2n + 3q$, then $e_q(n) \leq 14$.*
2. *If $x > 3$, every class is exposed. If $q \geq 29$ and $q^2 > 2n + 3q$, then $e_q(n) \leq \max(0, 4 - E_{\min}(n, q))$, where $E_{\min}(n, q)$ is the minimum of E_ξ over the q classes.*

Proof. (1) Block m contributes at least $\lfloor L_m/q \rfloor > L_m/q - 1$ zeros to the class, because its values $2(c-k_0)+1$ are L_m consecutive odd numbers. Level ℓ of the pole profile contributes fewer than $|I_\ell|/q+1$ poles, where $I_\ell := \{k : a_k \geq \ell\}$. Since $\sum_m L_m = N$ and $\sum_\ell |I_\ell| = N+4$, this gives $E > -10 - 4/q$, so $E \geq -10$. Then $L_a \geq E-4 \geq -14$, $\mathfrak{G} \geq E+1$ and $\mathfrak{C} \geq E+3$, and Theorem K gives $e_q(n) \leq 14$.

(2) The pole $k = k_0 \leq q-1$ of the class has $|d_{k_0}|/q \geq x/2 - 1 + 1/q > \frac{1}{2}$, so $\sigma = 1$ gives a bad pair. The bound follows from Theorem K with $L_a \geq E_{\min} - 4$, $\mathfrak{G} \geq E_{\min}+1$ and $\mathfrak{C} \geq E_{\min}+3$ (exposure is not needed for it: classes without bad pairs contribute nothing to B). □

Lemma 5.10 (Small primes)**.** *For every prime $q \geq 5$ and every n, $e_q(n) \leq 18\lfloor \log_q(2n+1) \rfloor$, and $e_3(n) \leq 19\lfloor \log_3(2n+1) \rfloor$.*

Proof. Put $\lambda_1 := \lfloor \log_q(2n+1) \rfloor$. For $k \neq \frac{n}{2}$ write $G_k(\varepsilon) = G_k(0)\prod_\beta (1+\varepsilon/\beta)^{e_\beta}$ with $\beta \in \frac{1}{2}\mathbb{Z} \setminus \{0\}$ and $|2\beta| \leq 2n+1$; so $v_q(\beta) \leq \lambda_1$, and Lemma 2.6 gives $v_q([\varepsilon^\lambda]\prod) \geq -\lambda\lambda_1$. For $k = \frac{n}{2}$, where the linear factor is 2ε and $G_k(0) = 0$, apply the same argument to $G_k(\varepsilon)/(2\varepsilon)$; since $H_{n/2} = \emptyset$, only $r_{3,n/2}$ matters there, and the count below is unchanged.

For $h \in H_k$ we have $|2h| < n$, and $X(k,h) = 24[\varepsilon^{a_k-1}]G_k(\varepsilon)(h-\varepsilon)^{-5}$ with $[\varepsilon^r](h-\varepsilon)^{-5} = \binom{r+4}{4}h^{-5-r}$. Hence $v_q(X(k,h)) \geq s_k - 8\lambda_1$ and $v_q(r_{3,k}) \geq s_k - \lambda_1$, where $s_k := v_q(G_k(0))$.

The valuation s_k is at least the numerator count minus the denominator count. In the numerator, the L_m values of block m are consecutive odd numbers, so their q-part is at least $\sum_e \lfloor L_m/q^e \rfloor$. In the denominator, for each level ℓ with $I_\ell \cap \mathbb{Z} = [\alpha, \beta']$, the product $\prod_{k' \in [\alpha,\beta'], k' \neq k} |k'-k|$ equals $(\beta'-\alpha)!/\binom{\beta'-\alpha}{k-\alpha}$ if $k \in [\alpha,\beta']$, and $|I_\ell|!\binom{\beta'-k}{|I_\ell|}$ (or the mirror image) otherwise; its q-part is at most $v_q(|I_\ell|!) + \lfloor \log_q n \rfloor$ by Kummer's bound. So

$$s_k \geq \sum_{e \geq 1} \Big[\sum_m \Big\lfloor \frac{L_m}{q^e} \Big\rfloor - \sum_\ell \Big\lfloor \frac{|I_\ell|}{q^e} \Big\rfloor \Big] - 4\lfloor \log_q n \rfloor.$$

Each bracket is an integer $> -6 - 4/q^e$, hence ≥ -6 (≥ -7 if $q=3$), and it vanishes once $q^e > n+1$. Hence $s_k \geq -10\lambda_1$ ($-11\lambda_1$ if $q=3$), and the lemma follows. □

Consequently $\sum_{q \leq Q} e_q(n) \log q \leq 19\,\pi(Q)\log(2n+1) = O(\sqrt{n})$ for $Q := \max(40, \sqrt{3n})$.

5.8. **The continuum function.** With $x = n/q$ and $y := k_0/q \in [0,1)$ the local model of a class is described by (x,y) as follows.

- *Poles.* The pole j exists if and only if $y + j \leq x$; its order is $a(\kappa)$ with $\kappa = (y+j)/x$, by the rule of §3.1 applied to $\min(\kappa, 1-\kappa)$.
- *Zeros.* The zero at $v = -s/2$ has multiplicity $\#\{m : \eta_m x < y + s/2 < (1-\eta_m)x\}$. The discrete end points are never hit, since $2k_0 + sq$ is odd while $2\eta_m n$ is even.
- *Bad pairs.* $d_k/q = y + j - \frac{x}{2}$, and (k,σ) is a bad pair if and only if $\sigma < 2|y + j - \frac{x}{2}|$; equality never occurs, since $|2k-n|$ is even and σq is odd.
- *Centre class.* This is the class $y \equiv \frac{x}{2} \pmod 1$, where the δ-zero sits on the pole at $v = -\lfloor x/2 \rfloor$.

As y runs over $[0,1)$ with x fixed, the local model changes only when y crosses a point $\equiv \gamma x + e \pmod 1$ with $\gamma \in \{b/40 : 0 \le b \le 40\}$ and $e \in \frac{1}{2}\mathbb{Z}$, or crosses $y \equiv 0$. These *events* come from pole existence ($\gamma = 1$), from the cells $b/40$ of the pole profile, from the block ends η_m, $1-\eta_m$, and from exposure and the centre ($\gamma = \frac{1}{2}$). The *candidate set* is $\mathcal{C} := \bigcup_{d=1}^{40} \frac{20}{d}\mathbb{Z}$.

Lemma 5.11 (Piecewise constancy)**.** *Two events $\gamma x + e$ and $\gamma' x + e'$ coincide modulo 1 only if $(\gamma - \gamma')x \in \frac{1}{2}\mathbb{Z}$, that is, $x \in \frac{20}{d}\mathbb{Z}$ with $d = 40|\gamma - \gamma'|$; and an event meets $y \equiv 0$ only if $\gamma x \in \frac{1}{2}\mathbb{Z}$. Hence the cyclic order of the events is constant on each open interval of $\mathbb{R} \setminus \mathcal{C}$. In particular the set $\mathcal{T}(x)$ of* continuum types *(the local model in each gap between consecutive events, together with the centre type) and the* continuum bound

$$f(x) := \max\bigl(0, -\min(\mathfrak{B}(\mathcal{T}(x)), \mathfrak{G}(\mathcal{T}(x))), -\mathfrak{C}(\mathcal{T}(x))\bigr)$$

are constant on each such interval.

Lemma 5.12 (Discrete types are continuum types)**.** *Let $q > 40$ be a prime with $q \nmid n/40$. Then $x = n/q \notin \mathcal{C}$, every local type realized by the q classes at (n, q) lies in $\mathcal{T}(x)$, and $f_n(q) \le f(x)$. Every prime $q > n/40$ satisfies $q \nmid n/40$.*

Proof. (i) If $x = 20m'/d$ with $d \le 40$, then $2(n/40)d = m'q$; since $q > 40 \ge d$ and q is odd, $q \mid n/40$. (ii) By the parity remarks above, a discrete class never sits on a zero event or an exposure event. A class on a pole-existence event or a cell event takes that event's one-sided state; since $x \notin \mathcal{C}$, no other event is at the same place, so its type is that of an adjacent gap. The class $k_0 \equiv \frac{n}{2}$ is the centre type. (iii) *Monotonicity.* Over fewer types, L_a, L_b, $\mathfrak{G}$ and $\mathfrak{C}$ can only increase. For L_κ: if the discrete P_0 equals the continuum P_0, some discrete type attains it, so the discrete kernel step is admissible with the same (j, c), and its levels are among the continuum ones; if the discrete P_0 is larger, then $\mathfrak{B}$(discrete) $\ge P_0$(discrete) $\ge P_0 + 1 \ge L_\kappa$(continuum). □

Lemma 5.13 (Periodicity)**.** *$E(x + 40, y) = E(x, y)$ for all (x, y), and $\mathcal{C} + 40 = \mathcal{C}$. Hence the residue level $f^0(x) := \max(0, 4 - \min_{\mathcal{T}(x)} E)$ is 40-periodic, and $f(x) \le f^0(x)$ for $x > 3$.*

Proof. Let $I_\ell^c := \{\kappa \in [0,1] : a(\kappa) \ge \ell\}$; these are the intervals $[0,1]$, $[\frac{1}{20}, \frac{19}{20}]$, $[\frac{1}{10}, \frac{9}{10}]$, $[\frac{3}{20}, \frac{17}{20}]$, of total length 3.4. Then

$$E = \sum_m \#\Bigl\{s \text{ odd} : y + \frac{s}{2} \in (\eta_m x, (1-\eta_m)x)\Bigr\} - \sum_{\ell=1}^{4} \#\{j \ge 0 : y + j \in x\, I_\ell^c\} + \delta.$$

Under $x \mapsto x+40$ every end point $40\eta_m$, $40(1-\eta_m)$ and $40\,\partial I_\ell^c$ is an integer, so each count grows by exactly $40(1-2\eta_m)$ or $40|I_\ell^c|$; since $\sum_m (1-2\eta_m) = 3.4 = \sum_\ell |I_\ell^c|$, the changes cancel, and the δ-point $y \equiv x/2$ moves by 20. The last claim is Lemma 5.9(2). □

5.9. **Theorem A.**

Theorem A. *As $n \to \infty$ through $40\mathbb{Z}$,*

$$\frac{1}{n}\log D_n \le E^* + o(1), \qquad E^* := \int_1^{60} f(x)\frac{\mathrm{d}x}{x^2} + \int_{60}^{\infty} f^0(x)\frac{\mathrm{d}x}{x^2} = 3.88084310467\ldots$$

Proof. We split the odd primes into five ranges.

(1) $q > n$: by Theorem C, $q \nmid D_n$.
(2) $\frac{n}{60} < q \le n$, $q \nmid \frac{n}{40}$: for $n \ge n_0$ we have $q > 40$ and $q^2 > 2n + 3q$, so $e_q(n) \le f(n/q)$ by Theorem K and Lemma 5.12. The function f is constant on the finitely many candidate intervals (a_i, b_i) of $[1, 60]$, and n/q is never an end point (the end points lie in $\mathcal{C}$). So by the prime number theorem $\sum f(n/q) \log q \le \sum_i f_i\big(\theta(n/a_i) - \theta(n/b_i)\big) = n \int_1^{60} f \, \frac{\mathrm{d}x}{x^2} + o(n)$.
(3) $\sqrt{3n} < q \le \frac{n}{60}$, $q \nmid \frac{n}{40}$: here $q^2 > 3n \ge 2n + 3q$, so $e_q(n) \le f^0(n/q)$ by Lemma 5.9(2) and Lemma 5.12. The function $f^0 \le 14$ is piecewise constant and 40-periodic. For any $X_2 > 60$, the prime number theorem on $[60, X_2]$ and Chebyshev's bound for $q \le n/X_2$ give at most $n \int_{60}^{\infty} f^0 \frac{\mathrm{d}x}{x^2} + 14.3\, n/X_2 + o(n)$.
(4) $q \mid \frac{n}{40}$, $q > \sqrt{3n}$: there are at most $\log_2 n$ such primes, each with $e_q(n) \le 14$ (Lemma 5.9(1)); they contribute $O(\log^2 n)$.
(5) $q \le \max(40, \sqrt{3n})$: $O(\sqrt{n})$, by Lemma 5.10.

Letting $X_2 \to \infty$ proves the theorem. The value of E^* is computed in Table 1. □

TABLE 1. The value of E^*. The step function f is computed exactly on each candidate interval. The periodic tail is computed from the exact values f_i^0 of f^0 on the 980 intervals (a_i, b_i) of the period $[60, 100)$ as $\sum_i f_i^0(\psi(b_i/40) - \psi(a_i/40))/40$, with the digamma function ψ, $\psi(z) - \psi(w) = \sum_{j \ge 0}(\frac{1}{j+w} - \frac{1}{j+z})$.

piece	value	exact value	computation
$\int_1^4 f \, \mathrm{d}x/x^2$	2.55	51/20	77 intervals
$\int_4^{20} f \, \mathrm{d}x/x^2$	1.03686695249…	2989163/2882880	392 intervals
$\int_{20}^{60} f \, \mathrm{d}x/x^2$	0.19208902168…	see below	980 intervals
$\int_{60}^{\infty} f^0 \, \mathrm{d}x/x^2$	0.10188713049…	–	one period, digamma sum
E^*	3.88084310467…		

The exact value of the third piece is

$$\int_{20}^{60} f \, \frac{\mathrm{d}x}{x^2} = \frac{52635044926839457 63}{27401381122591476000}.$$

Remark 5.14. (1) The function f^0 has mean 6.0753 and maximum 10 over one period. The tail is not delicate: using only $f^0 \le 10$ beyond 60 gives $E \le 3.9456$. (2) On $[1, 3.2)$, f is the step function with values $0, 1, 2, 3, 4, 5, 4, 2, 3, 4, 5$ and breaks at $\frac{20}{19}, \frac{10}{9}, \frac{20}{17}, \frac{5}{4}, \frac{4}{3}, \frac{20}{13}, 2, \frac{20}{9}, \frac{40}{17}, \frac{20}{7}$; its integral there against $\mathrm{d}x/x^2$ is 181/80. (3) The kernel step with $j \ge 1$ (Lemma 5.5) is used only on $[2, \frac{20}{9})$, with $j = 2$ and $c = -\frac{5}{4}$; the bounds $\mathfrak{G}$ and $\mathfrak{C}$ are never binding on $[1, 60]$. (4) Where the saving comes from: with the same tail, the residue level $4 - E$ alone gives the exponent 5.0818; the class lemma with $\kappa = 0$ gives 4.3593; adding the identity with $\kappa = 1$ gives 3.9843; the one-step kernel gives $E^* = 3.8808$.

Remark 5.15 (Evidence and a second proof). The following checks are not used in the proof. The bound of Theorem K was compared with the true exponents $e_q(n)$, computed from exact odd denominators by independent programs, at 11665 pairs (n, q) for 22 values of n up to 200000 (including every prime at $n = 20000$), and at

1431 further primes for $n = 480, \dots, 20000$: there was no violation, and the bound is attained at 11560 of the 11665 pairs. Of the 1449 head intervals, the true exponent attains f on 775, which carry 97.3% of $\int_1^{60} f\,\mathrm{d}x/x^2$; so E^* is essentially sharp for this method. The class lemma was tested as an exact identity (valuation and leading digit modulo q) on 19827 class instances. A second, independent proof of a bound of the same kind, which chooses the kernel subtractions by exact Gaussian elimination and uses a first-order bound beyond $x = 40$, gives the exponent 3.8801844 (margin -0.1265142) and was compared prime by prime (19063 comparisons, no violation).

6. The archimedean size (Theorem G)

Let $a(z)$ be the continuum pole profile on $[0,1]$: $a(z) = 1, 2, 3$ for $\min(z, 1-z)$ in $[0, \frac{1}{20})$, $[\frac{1}{20}, \frac{1}{10})$, $[\frac{1}{10}, \frac{3}{20})$, and $a(z) = 4$ otherwise. Define the potential

$$U(\kappa) := \sum_{m=1}^{6} \int_{\eta_m}^{1-\eta_m} \log|z-\kappa|\,\mathrm{d}z - \int_0^1 a(z)\log|z-\kappa|\,\mathrm{d}z \qquad (\kappa \in [0,1]).$$

With $F(x) := x\log|x| - x$ this is a finite sum $U(\kappa) = \sum_j w_j F(p_j - \kappa)$ of 18 terms with integer weights w_j and break points $p_j \in \frac{1}{40}\mathbb{Z} \cap [0,1]$.

Lemma 6.1 (G1, Cauchy's estimate)**.** *We have $|r_{i,k}| \le 2^{-i} M_k$, where*

$$M_k := \max_{|\varepsilon|=1/2} |R_n(-k+\varepsilon)|.$$

Proof. G_k is holomorphic on $|\varepsilon| < 1$, and Cauchy's estimate on $|\varepsilon| = \frac{1}{2}$ [1] gives $|[\varepsilon^\lambda]G_k| \le 2^{\lambda - a_k} M_k$. Take $\lambda = a_k - i$. □

Lemma 6.2 (G2, triangle inequality)**.**

$$\log M_k \le \log(|n-2k|+1) + a_k\log 2 + \sum_m \sum_{c\in C_m} \log\Big(\Big|c-k+\frac{1}{2}\Big| + \frac{1}{2}\Big) - \sum_{k'\ne k} a_{k'} \log\Big(|k'-k| - \frac{1}{2}\Big).$$

Lemma 6.3 (G3, sums and integrals)**.** *Let $f(y) := \log|y-k|$ and $C_3 := 2(1-\log 2) + \pi^2/72 + 2 = 2.750783\ldots$*

1. *For an integer interval $[\mathrm{lo}, \mathrm{hi}]$ of length L, $\sum_{c=\mathrm{lo}}^{\mathrm{hi}} \log(|c-k+\frac{1}{2}| + \frac{1}{2}) \le \int_{\mathrm{lo}}^{\mathrm{hi}+1} f + C_3 + \log(2L)$.*
2. *$\sum_{k'\ne k} a_{k'} \log(|k'-k| - \frac{1}{2}) \ge \int_{-1/2}^{n+1/2} \tilde{a}_n f - 8(1+\log n)$, where $\tilde{a}_n := a_{k'}$ on $[k'-\frac{1}{2}, k'+\frac{1}{2})$.*

Proof. (1) Split $\log(|u|+\frac{1}{2}) = \log|u| + \log(1+1/(2|u|))$, $u = c-k+\frac{1}{2}$. The second part is at most $\sum 1/|2(c-k)+1| \le 2+\log(2L)$. For the first part use the midpoint rule on the cells $[c, c+1]$: each of the two cells adjacent to k contributes at most $1-\log 2$, and a cell at distance $d \ge 1$ at most $1/(24d^2)$, since $|f''| \le 1/d^2$ there. (2) Use $\log(d-\frac{1}{2}) \ge \log d - 1/d$ and $\sum_{k'\ne k} a_{k'}/|k'-k| \le 8H_n$. On each cell not containing k, f is concave, so its midpoint value is at least its mean. The omitted cell of k contributes $-a_k(1+\log 2) \le 0$ to the integral. □

Lemma 6.4 (G4–G5, scaling and discretization)**.** *Substituting $y = nz$ gives $\int_{\alpha n}^{\beta n} f = (\beta-\alpha)n\log n + n\int_\alpha^\beta \log|z-k/n|\,\mathrm{d}z$, and similarly for $\tilde a_n$. The discrete profile $\tilde a_n(nz)$ differs from $a(z)$ only on 8 intervals (the six interior jumps and the two end half-cells), each of length at most $1/(2n)$, and there by 1; hence $n\big|\int(\tilde a_n(nz)-a(z))\log|z-\kappa|\,\mathrm{d}z\big| \le 4(1+1/(2n)+\log 4n)$.*

Proposition 6.5 (G6)**.** *For all $0\le k\le n$ and $n\ge 40$,*

$$\log M_k \le n\,U(k/n) + \log(n+1) + 14\log n + 37.44.$$

Proof. Combine Lemmas 6.2–6.4. The two $\log n$ terms of the scaling combine to $(N-\sum_k a_k)\log n = -4\log n$, and the constants add up to $37.386 + 2/n \le 37.44$ (with $\sum_m \log(2-4\eta_m) = 0.5635$ from the $\log(2L)$ terms). □

Lemma 6.6 (G7, certified maximum)**.** $0.7067022241 \le \max_{[0,1]} U \le U^* := 0.70670223$*; the maximum is attained at $\kappa^* = 0.1067347196\ldots$ and at $1-\kappa^*$.*

The proof is a computer-assisted certification, described in §9.1.

Theorem G. *For every $n\in 40\mathbb{Z}_{>0}$,*

$$\log\max(|\rho_n|,|Z_n|) \le U^* n + 14\log n + 2\log(n+1) + 47.5.$$

Proof. By (3.3) and Lemma 6.1,

$$\begin{aligned}|\rho_n| &\le \sum_{k,i}(i)_4 2^{-i} M_k \sum_{h\in H_k}|h|^{-i-4} \le 16\sum_{i=1}^{4}(i)_4(1-2^{-i-4})\zeta(i+4)\sum_k M_k\\ &\le 21514\,(n+1)\max_k M_k,\end{aligned}$$

using $\sum_h |h|^{-s} \le 2^s(1-2^{-s})\zeta(s)$ for distinct half-integers of one sign. Also $|Z_n| = 46080|\sum_k r_{3,k}| \le 5760(n+1)\max_k M_k$. By Proposition 6.5, Lemma 6.6 and $\log 21600 + 37.44 \le 47.5$, the bound follows. □

Remark 6.7. The true growth of $\max(|\rho_n|,|Z_n|)$ is about $e^{0.684n}$, smaller than e^{U^*n}: the largest residues sit in a stretch of pole order 3 where their signs alternate, and they cancel (measured: 0.6837 at $n = 96000$). A saddle-point argument could lower U^* by about 0.023; this is not needed.

7. Nonvanishing (Theorem N)

Let

$$I := \{40m :\ 38m-1 \text{ prime}\} = \{40, 120, 160, 240, 400, 600, 640, 720, 840, 960, \ldots\}.$$

By Dirichlet's theorem I is infinite, and for $n\in I$ the number $\ell := \frac{19}{20}n - 1 = 38m - 1 \ge 37$ is prime.

Theorem N. *Let $n\in 40\mathbb{Z}_{>0}$ and suppose that $\ell := \frac{19}{20}n - 1$ is prime. Then*

$$v_\ell(\rho_n) = -1, \qquad v_\ell(Z_n)\ge 0, \qquad \ell\rho_n \equiv \frac{6864}{7}\cdot\frac{r_{1,0}}{\ell^4} \pmod{\ell},$$

where $u_0 := r_{1,0}/\ell^4$ is an ℓ-unit.

Proof. Throughout, $\ell \geq 37$, $2n < 3\ell$ and $\ell^2 > 2n$.

Size facts. (S1) $|k' - k| \leq n < 2\ell$, so $\ell \mid k' - k$ if and only if $k' - k = \pm\ell$. (S2) The numerator values $2(c-k)+1$ are odd with absolute value at most $2n-1 < 3\ell$, so ℓ divides one only if it equals $\pm\ell$, that is, $c = k + \frac{\ell-1}{2}$ or $c = k - \frac{\ell+1}{2}$, and then with valuation 1. (S3) $n - 2k$ is even and $|n - 2k| < 2\ell$, so $\ell \nmid n - 2k$ unless $k = \frac{n}{2}$. (S4) For $h \in H_k$, $|2h| \leq n - 1 < 2\ell$, so $\ell \mid 2h$ if and only if $2h = \pm\ell$, which forces $|k - \frac{n}{2}| \geq \frac{19}{40}n$, that is, $k \leq \frac{n}{40}$ or $k \geq \frac{39}{40}n$. (S5) Every block contains the window $[\frac{19}{40}n - 1, \frac{21}{40}n]$, and the numbers $(i)_4$ and 46080 are ℓ-units.

N1 (local data). Write $s_k := v_\ell(G_k(0))$. (a) *End poles* $0 \leq k \leq \frac{n}{20} + 1$. The value $c = k + \frac{19}{40}n - 1$ lies in the window, hence in all six blocks, and gives $2(c-k)+1 = \ell$; the other branch $c = k - \frac{19}{40}n < 0$ lies in no block. The pole partner $k + \ell$ has order 2 for $k \in \{0, 1\}$ and order 1 for $k \geq 2$. So $s_0 = s_1 = 6 - 2 = 4$ and $s_k = 5$ for $2 \leq k \leq \frac{n}{20} + 1$. By Lemma 2.6, $v_\ell(r_{i,k}) \geq s_k - (a_k - i) \geq 3$. The mirror poles obey the same bounds by Lemma 3.1(c). (b) *Middle poles* $\frac{n}{20} + 1 < k < \ell$. Here the pole partners $k \pm \ell$ lie outside $[0, n]$, so only numerator factors ($e_\beta \geq 0$) are divisible by ℓ, and Lemma 2.6 gives $v_\ell(r_{i,k}) \geq 0$ (at $k = \frac{n}{2}$ apply it to $G_k(\varepsilon)/(2\varepsilon)$). (c) *Consequence.* Every c_i is ℓ-integral, so $v_\ell(Z_n) \geq 0$.

N2 (two terms). Use $\rho_n = -2\sum_{k<n/2}\sum_{h\in H_k} X(k,h)$. By N1, the terms with $\ell \nmid 2h$ are ℓ-integral. For $k < \frac{n}{2}$ the remaining terms have $2h = -\ell$, which requires $k \leq \frac{n}{40}$ by (S4); conversely $h = -\frac{\ell}{2} \in H_k$ for all $k \leq \frac{n}{40}$. There $a_k = 1$, and $X_k := X(k, -\frac{\ell}{2}) = 24 r_{1,k}(-\frac{\ell}{2})^{-5} = -768\, r_{1,k}/\ell^5$ has valuation $s_k - 5$, which is -1 for $k \in \{0, 1\}$ and ≥ 0 otherwise. Hence $\rho_n \equiv -2(X_0 + X_1) \pmod{\mathbb{Z}_{(\ell)}}$, that is, $\ell\rho_n \equiv 1536\,(r_{1,0} + r_{1,1})/\ell^4 \pmod{\ell}$.

N3 (the ratio). Since $r_{1,k} = G_k(0) = (n - 2k)\,\mathrm{Num}(-k)/\prod_{k'\neq k}(k' - k)^{a_{k'}}$ for $k \in \{0, 1\}$, where Num is the product of the six blocks, the numerator products telescope blockwise, and the ratio of the pole products only involves the jumps of the profile:

$$\Theta := \frac{r_{1,1}}{r_{1,0}} = \frac{n-2}{n}\prod_{m=1}^{6}\frac{2\eta_m n - 1}{2(1-\eta_m)n - 1}\cdot\frac{-n\cdot\frac{17}{20}n\cdot\frac{9}{10}n\cdot\frac{19}{20}n}{(\frac{n}{20}-1)(\frac{n}{10}-1)(\frac{3}{20}n-1)}.$$

Since $\frac{19}{20}n = \ell + 1$, we have $n \equiv \frac{20}{19} \pmod{\ell}$, and then $\frac{n-2}{n} \equiv -\frac{9}{10}$, the block product is $\equiv (-\frac{7}{8})(-\frac{6}{7})^2(-\frac{5}{6})(-\frac{4}{5})(-\frac{3}{4}) = \frac{9}{28}$, and the pole factor is $\equiv \frac{5}{4}$. Hence $\Theta \equiv \theta_0 := -\frac{81}{224}$ and $1 + \theta_0 = \frac{143}{224} = \frac{11\cdot 13}{2^5\cdot 7}$. All factors involve only the primes $2, 3, 5, 7, 11, 13, 17, 19 < 37 \leq \ell$, so the reduction is legitimate.

Conclusion. $u_0 = r_{1,0}/\ell^4$ is an ℓ-unit because $s_0 = 4$, and $\ell\rho_n \equiv 1536(1+\theta_0)u_0 = \frac{6864}{7}u_0 \not\equiv 0 \pmod{\ell}$, since $\frac{6864}{7} = \frac{2^4\cdot 3\cdot 11\cdot 13}{7}$ is an ℓ-unit. Hence $v_\ell(\rho_n) = -1$. □

Remark 7.1. The constants were recomputed exactly by a separate program. The leading residues $\ell\rho_n \bmod \ell$ at the ten $n \in I$ with $n \leq 960$ are 24, 64, 54, 26, 101, 510, 271, 224, 704, 434, as predicted; Theorem N was also checked ℓ-adically at all 334 values $n \in I$ with $n \leq 60000$.

8. Proof of the Main Theorem

Suppose that $\zeta_2(7) = u/b$ with $u \in \mathbb{Z}$ and $b \in \mathbb{Z}_{\geq 1}$. For $n \in I$ choose $e_n \geq 0$ with $A_n := 2^{e_n} D_n \rho_n \in \mathbb{Z}$ and $B_n := 2^{e_n} D_n Z_n \in \mathbb{Z}$, and put $N_n := bA_n + uB_n \in \mathbb{Z}$. By Lemma 3.2, $N_n = b\,2^{e_n} D_n\, S_n$ in $\mathbb{Q}_2$.

Step 1 ($N_n \neq 0$). Let $n \in I$ with $\ell := \frac{19}{20}n - 1 > b$. By Theorem N, $v_\ell(bA_n) = v_\ell(D_n) - 1$, and $v_\ell(uB_n) \geq v_\ell(D_n)$ (or $uB_n = 0$). So $v_\ell(N_n) = v_\ell(D_n) - 1 < \infty$ and $N_n \neq 0$. (This is condition (2) of [16, Lemma 2.2] with $\ell(n) = \frac{19}{20}n - 1$.)

Step 2 (Liouville). A non-zero integer N satisfies $1 \leq |N| \cdot |N|_2$. Here $|N_n| \leq (b + |u|)\, 2^{e_n} D_n \max(|\rho_n|, |Z_n|)$ and $|N_n|_2 \leq 2^{-e_n} 2^{-v_2(S_n)}$, since D_n is odd. Hence

$$1 \leq (b + |u|)\, D_n \max(|\rho_n|, |Z_n|)\, 2^{-v_2(S_n)} \qquad \text{for all } n \in I \text{ with } \tfrac{19}{20}n - 1 > b. \tag{8.1}$$

Step 3. By Theorem A (valid for all $n \in 40\mathbb{Z}$, hence along I), Theorem G and (4.1), the logarithm of the right-hand side of (8.1) is at most

$$\log(b + |u|) + (E^* + U^* - 6.8 \log 2)\, n + o(n) = -0.1258555\ldots \cdot n + o(n) \longrightarrow -\infty.$$

Since I is infinite, this contradicts (8.1). Hence $\zeta_2(7)$ is irrational. □

Only the asymptotic inequality $E^* + U^* < 6.8 \log 2$ is used: given E^*, any growth exponent below $6.8 \log 2 - E^* = 0.8325577$ suffices, and given U^*, any denominator exponent below $6.8 \log 2 - U^* = 4.0066986$. The $o(n)$ of Theorem A is ineffective, and we claim no irrationality measure. Table 2 lists the margins obtained with the various bounds.

TABLE 2. Margins $E + g - 6.8 \log 2$ per n, where $6.8 \log 2 = 4.7134008$ and $U^* = 0.70670223$. The last two rows are numerical and not used in the proof.

growth g	denominator exponent E	margin	status
U^*	$E^* = 3.88084310$ (Theorem A)	-0.1258555	proved
U^*	3.88018442 (second proof, Remark 5.15)	-0.1265142	proved
U^*	3.9240892 (kernel level on $[1, 20]$, residue level on $[20, 140]$, 10/140 beyond)	-0.0826094	proved; Lean uses $g = 0.71$, $E = 3.99$
U^*	3.9308 (without Lemma 5.5)	-0.0759	proved
U^*	3.9843 (class lemma with $\kappa = 0, 1$ only)	-0.0224	proved
≈ 0.684	$\approx 3.87 \pm 0.02$ (true asymptotic value)	≈ -0.16	numerical
true	true D_n, $480 \leq n \leq 6000$	-0.23 to -0.18	numerical

Remark 8.1 (Finite n). For $n = 480$, 640, 960, 1440, 2000 and 6000, the quantity $\frac{1}{n} \log\big(D_n \max(|\rho_n|, |Z_n|) 2^{-v_2(S_n)}\big)$ computed from the exact data is -0.2296, -0.2161, -0.2226, -0.1802, -0.1829 and -0.1768. Finite n is far from the asymptotic regime for the prime sums: the large-prime part of $\frac{1}{n} \log D_n$ grows from about 3.3 ($n = 480$) to 3.70 ($n = 6000$), towards E^*, while the small-prime part decays like $n^{-1/2}$. These numbers are not used in the proof.

9. The computer-assisted steps

All computations of the informal proof were done in exact rational arithmetic (Python `fractions`) or in interval arithmetic with outward rounding (`mpmath.iv` [20]), and every computer-assisted step was redone by an independent implementation written by a different agent.

9.1. **The maximum of U (Lemma 6.6).** Each term $w_j F(p_j - \kappa)$ is monotone in κ on $[-1, 1]$, and convex or concave on each interval between consecutive break points (since $F''(z) = 1/z$). Three certifications were made.

(1) A monotone box bound, a centred first-order bound on boxes without break points, and bisection from the grid $\frac{1}{40}\mathbb{Z}$: 152 boxes, run twice, once in `mpmath.iv` at 120 bits with outward rounding and once in exact rationals with the program's own enclosures of log (from the series of artanh with explicit remainders). The negative control 0.70670222 fails, as it must.
(2) An independent program bounds the convex terms by chords and the concave terms by tangents, with exact rationals and correctly rounded decimal logarithms, on 109 boxes; it finds $\max U = 0.7067022242140760\ldots$
(3) The Lean proof certifies the weaker bound $\max U \le 0.71$ by chords and tangents on 42 boxes with end points in $\frac{1}{80}\mathbb{Z}$ (worst box bound 0.708146), with rational enclosures of log checked in Lean (§10).

9.2. **The value of E^* (Theorem A).** The function f of Lemma 5.11 is evaluated in exact rational arithmetic on each of the 1449 candidate intervals of $[1, 60]$, from the continuum types of the interval (Lemma 5.11); for each interval the program records f and the kernel that certifies it. The residue level f^0 is computed on one period $[20, 60]$ (980 intervals), and the tail beyond 60 is the digamma sum of Table 1. An independent implementation, which formulates the levels through residue moments modulo $2^{127} - 1$ and uses its own event set, reproduces all 1449 interval values exactly, and the three head integrals were re-summed exactly (51/20, 2989163/2882880 and 5263504492683945763/27401381122591476000). The formal proof re-checks, in the Lean kernel, a table of 469 kernel-level rows on $[1, 20)$ and 980 residue-level rows on $[20, 60)$ (extended by periodicity to $[20, 140)$), and the resulting bound $3.9240891610\ldots \le 3.99$ (§10).

9.3. **Other numerical inputs.** The constants of Theorem N ($\theta_0 = -\frac{81}{224}$ and $1536(1 + \theta_0) = \frac{6864}{7}$), of Theorem G ($21513.3\ldots$, 5760 and the constant of Proposition 6.5) and of Table 2 are hand computations that were rechecked in exact arithmetic.

9.4. **Code and data.** The exact reference engines of the research (the computation of $r_{i,k}$, ρ_n, Z_n and of the levels of Section 5), the interval data of f on $[1, 20]$ and the interval data of f^0 on one period (the first, second and fourth pieces of Table 1) are included, together with an exact-arithmetic mirror of the Lean definitions and an independent audit script, in the directory `python/` of the formalization repository (§10.6); the interval data of f on $[20, 60]$ and the other certification programs of this section are not. The formal proof does not depend on any of these programs.

10. Formal verification

The Main Theorem is formalized in Lean 4 [21] with Mathlib [19]. The development extends the formalization [11] of the statement that at least one of $\zeta_2(7)$, $\zeta_2(9)$ is irrational: it keeps that formalization's trusted definitions, its shared proofs (the convergence of J_s, the translation formula, Lai's criterion, the Δ-calculus) and its proof of the prime number theorem, and adds the namespace `Zeta2.Z7` (directory

`Zeta2Lean/Z7/`) for the present proof. Lean names are quoted verbatim; in displayed Lean code the Unicode symbols of the source are typeset as mathematical symbols.

10.1. **The statement.** The main theorem (file `Zeta2Lean/Z7/Main.lean`) is

```
theorem Zeta2.Z7.zeta2_7_irrational :
    (∀ s : ℕ, HasVolkenborn (halfPow s) (J s)) ∧ ¬ ∃ q : ℚ, zeta2 7 = q
```

Here `q` is coerced from $\mathbb{Q}$ to Mathlib's field $\mathbb{Q}_2$ `[2]` of 2-adic numbers, so the second conjunct says that `zeta2 7` is irrational in the sense of §1.2. The first conjunct says that for every s the Riemann sums of $(x+\frac{1}{2})^{-s}$ converge in $\mathbb{Q}_2$ to `J s`; it is there because `J s` is defined with Mathlib's `limUnder`, which returns an unspecified value when the limit does not exist.

10.2. **The trusted definitions.** The statement uses six definitions (namespace `Zeta2`, file `Zeta2Lean/Defs.lean`, with the namespaces `Filter`, `Topology`, `Finset` and `PowerSeries` opened). They are verbatim identical to those of the registered formalization [11, 24]:

```
def volkenbornSum (f : ℕ → ℚ_[2]) (N : ℕ) : ℚ_[2] :=
  ((2 : ℚ_[2]) ^ N)⁻¹ * ∑ x ∈ range (2 ^ N), f x

def HasVolkenborn (f : ℕ → ℚ_[2]) (I : ℚ_[2]) : Prop :=
  Tendsto (volkenbornSum f) atTop (𝓝 I)

def volkenborn (f : ℕ → ℚ_[2]) : ℚ_[2] :=
  limUnder atTop (volkenbornSum f)

def halfPow (s x : ℕ) : ℚ_[2] :=
  ((x : ℚ_[2]) + 2⁻¹)⁻¹ ^ s

def J (s : ℕ) : ℚ_[2] :=
  volkenborn (halfPow s)

def zeta2 (s : ℕ) : ℚ_[2] :=
  J (s - 1) / (((s : ℚ_[2]) - 1) * 2 ^ s)
```

Here $\mathcal{N}\,I$ is the neighbourhood filter of I, so `HasVolkenborn f I` says that the Riemann sums (2.1) converge to I. Hence `J s` is the integral J_s of (2.2), and `zeta2 7` $= J_6/768$. All other definitions (the rational function, ρ_n, Z_n, the local models, the certificate tables, the rates) are internal to the proof: if one of them differed from the text, the formal theorem would still hold as stated.

10.3. **What is cited.** The identification `zeta2 7` $= J_6/768 = \zeta_2(7) = L_2(7,\omega^{-6})$ is Lemma 2.8 of [17] (Lemma 2.3). It is cited and not formalized, since Mathlib does not yet contain the Kubota–Leopoldt L-functions (a Lean 3 formalization of p-adic L-functions exists [22], but it is not part of Mathlib). It is needed only to read the formal theorem as a statement about the Kubota–Leopoldt value, not in its proof. Since 768 is a non-zero rational number, given this identification the formal theorem is equivalent to the Main Theorem, and to the analogous statement for every normalization that differs by a non-zero rational factor.

10.4. **Structure of the formal proof.** The formalization follows a blueprint. Each ingredient of the informal proof is a Lean proposition `Stmt_X` (file `Zeta2Lean/Z7/Statements.lean`), proved in its own file of `Zeta2Lean/Z7/Proofs/` with the statements it depends on as hypotheses. The file `Zeta2Lean/Z7/Assembly.lean` derives the main theorem, for any rates g, δ with $g + \delta < 6.8 \log 2$, from the top-level statements, and `Main.lean` connects all proofs. Table 3 gives the correspondence. The formal proof differs from the text in the following points.

(1) Sign and indexing: $n = 40u$, and the Lean objects `rho`, `Zc` are $-\rho_n$, $-Z_n$ (the integrand is $-R_n'''(t + \frac{1}{2} - \frac{n}{2})$).
(2) Rates: the growth rate is $g = 0.71 \geq U^*$ and the denominator rate is $\delta = 3.99$, with the margin $0.71 + 3.99 = 4.70 < 6.8 \log 2$ (`margin7`, from Mathlib's bound $\log 2 > 0.6931471803$).
(3) Denominators: the formal Theorem A uses the kernel-level bound on $[1, 20)$ (469 rows), the residue level on $[20, 140)$ (3×980 rows, by periodicity), the bound $e_q(n) \leq 10$ for $n/q \geq 140$ (that is, $E_\xi \geq -6$), Lemma 5.9(1) for $q \mid u$, and Lemma 5.10 for $q \leq 1000$ or $q^2 \leq 2n + 3q$. The table integral is $3.9240891610\ldots$, and the kernel-checked bound is ≤ 3.99. The certificate statements assume $q > 1000$.
(4) Growth: the residue bound $|r_{i,k}| \leq C(u+1)^B e^{nU(k/n)}$ is proved without complex analysis and without integrals: a weighted norm on truncated power series replaces Cauchy's estimate, and exact telescoping sums of $F(z) = z \log |z| - z$ replace Lemmas 6.3–6.4, with weaker but sufficient polynomial factors. The maximum of U is certified as in §9.1(3).
(5) Nonvanishing: Dirichlet's theorem is Mathlib's `Nat.frequently_atTop_prime_and_modEq`, and Lai's criterion is used in the form "$|a_{n,j}| \leq B_n$ for all j, $B_n |L_n|_2 \to 0$ and $L_n \neq 0$ for infinitely many n", proved in [11].

In the course of the formalization one small gap in the informal proof was found and repaired: in Lemma 5.10 the factorization of G_k fails at the centre $k = \frac{n}{2}$, where $G_k(0) = 0$; the text above already contains the repair (work with $G_k(\varepsilon)/(2\varepsilon)$).

10.5. **Trust base and checks.**

- *Lean and Mathlib.* Lean `v4.35.0-rc2` (toolchain `leanprover/lean4:v4.35.0-rc2`) and Mathlib at commit `065356127b`, as in [11].
- *Axioms.* `#print axioms` applied to `Zeta2.Z7.zeta2_7_irrational` lists exactly `propext`, `Classical.choice` and `Quot.sound`. The script `scripts/audit.sh` finds no `sorry`, no `axiom` declaration and no `native_decide`, `implemented_by` or `extern`.
- *Numerical certificates.* The denominator table and its integral bound are checked by `decide +kernel`, that is, by reduction in the kernel; the bound $\max U \leq 0.71$ is proved from rational enclosures of log whose side conditions are checked by `norm_num`. The Python programs that generated these certificates need not be trusted.
- *Vendored code.* The prime number theorem is proved from Lean files of open Mathlib pull requests by T. Tao, adapted from the PrimeNumberTheoremAnd project [26] (their pull-request descriptions state that AI tools were used in writing or golfing the code). They are included with their headers and authors; apart from the imports and the module header, their code is

TABLE 3. The formalization. Lean statements are in the namespace `Zeta2.Z7`, and proof files in `Zeta2Lean/Z7/Proofs/`.

this paper	Lean statement	proof files
Lemmas 3.1, 3.2	`Stmt_PF7`, `Stmt_CoeffVanish7`, `Stmt_TranslationNeg`, `Stmt_GenLinearFormZ`, `Stmt_L7`	`PartialFractions7`, `CoeffVanish7`, `TranslationNeg`, `GenLinearFormZ`, `LinearForm7`
Theorem V	`Stmt_Valuation7`	`IntegrandTaylor7`, `Valuation7`
Theorem C	`Stmt_BigPrimes7`	`BigPrimes7`
Lemmas 5.3, 5.4, 5.5, 5.8	`Stmt_ClassLemma7`, `Stmt_KernelIdentity7`, `Stmt_FarPart7`, `Stmt_PerPole7`	`ClassLemma7`, `LocalExpansion7`, `KernelIdentity7`, `FarPart7`, `PerPole7`
Theorem K, Lemma 5.9(1)	`Stmt_ThmK7`, `Stmt_Emin7`	`ThmK7`, `Emin7`
Lemma 5.10	`Stmt_SmallPrimes7`	`SmallPrimes7`
Lemmas 5.11–5.13, Table 1	`Stmt_HeadCert7`, `Stmt_ResCert7`, `Stmt_ExpBounds7`	`HeadCert7*`, `ResCert7*`, `ExpBounds7`; table `Zeta2Lean/Z7/DenTable`
Theorem A	`Stmt_Denominators7 delta7`	`Denominators7`
Lemmas 6.1–6.6, Theorem G	`Stmt_ResidueBound7`, `Stmt_MaxU7 g7`, `Stmt_Growth7 g7`	`ResidueBound7`, `MaxU7`, `Growth7`
Theorem N	`Stmt_AuxPrime7`, `Stmt_Nonvanishing7`	`AuxPrime7`, `NonvanishingLS7`
Section 8	`main7_of_stmts`, `margin7`	`Zeta2Lean/Z7/Assembly`
Lemmas 2.1, 2.4, 2.5; PNT	`Stmt_Translation`, `Stmt_Delta`, `Stmt_DeltaFun`, `Stmt_Criterion`, `Stmt_JConv`, `PNT_Stmt` (namespace `Zeta2`)	`Zeta2Lean/Proofs/*`, `Zeta2Lean/Cited/PNT`, shared with [11]
Main Theorem	`zeta2_7_irrational`	`Zeta2Lean/Z7/Main`

identical to that of the pull requests. Being Lean proofs checked by the kernel, they add nothing to the trust base.

- *Kernel replay.* Lean's `leanchecker` replayed all 70 modules of the import closure of `Zeta2Lean.Z7.Main` at the development commit of §10.6, one module at a time, without error (at most about 13 GB of memory per module, for the certificate data).
- *Mirror and audit.* An exact-arithmetic Python mirror of the Lean definitions checks every statement numerically against the research programs, and an independent audit script, which shares no code with the mirror, tested the statements on about 700,000 instances, with no failure. These checks concern the faithfulness of internal definitions and are not part of the trust base.

10.6. **How to check.** The formalization is published at `https://github.com/gmDevi/zeta2-7-lean`, commit `e26813f04b3e84e136c88c00753567e1b442d5db`. It is the development described here, repackaged for the Palomar registry: every file uses the Lean module system, unused files were removed, and the heavy certificates are checked in small pieces so that every kernel check fits in 32 GB; no statement changed. This commit is registered in the Palomar registry of Lean-verified results as entry PALOMAR-2026-09-30-000018 [25]. With `elan` installed, the commands

```
lake exe cache get
lake build
lake env lean Zeta2Lean/Z7/Main.lean
bash scripts/audit.sh
bash scripts/kernels.sh Zeta2Lean.Z7.Main
```

fetch the compiled Mathlib, build the project, print the axioms of the main theorem, run the census of `sorry` and forbidden constructs, and replay the import closure of the main theorem with `leanchecker`, one module at a time.

11. AI use and verification

11.1. **How this work was produced.** Apart from the published results it cites and the vendored third-party Lean files for the prime number theorem [26], everything in this paper was produced by AI agents, namely Anthropic Claude models, under the direction of the author, in September 2026. The author posed the research question, directed and supervised the work, and is responsible for this paper. In detail:

(1) *Research.* An orchestrating agent planned the research and launched subagents. Research agents explored families of rational functions with exact rational arithmetic; the construction of Section 3 was found by a genetic search over pole-order ramps and nested blocks, guided by numerical estimates of the four exponents.
(2) *Proofs.* Separate agents proved the ingredients: one the denominator bound (Theorems C, K, A), one a second, independent version of it, and one the growth, the 2-adic size and the nonvanishing (Theorems G, V, N).
(3) *Verification of the informal proof.* Each ingredient was re-derived by a separate adversarial verifier agent with its own code. An assembly agent combined the ingredients, recomputed every number of the margin table, and wrote an AI-written referee report on the assembled proof. The verifiers found errors of detail (for instance the inadmissible kernels of Remark 5.7, and the interval of the local points in Lemma 5.2); all were corrected, and none affected the conclusions. Their existence means that the text may still contain further slips.
(4) *Formalization.* An architect agent wrote the Lean definitions and the statements of all lemmas. Prover agents wrote the proofs against these frozen statements and reported problems instead of changing them. An auditor agent checked the definitions and statements for faithfulness to the informal proof and for soundness (no vacuous statements, no loopholes through junk values such as $x/0 = 0$ or truncated subtraction), with independent exact-arithmetic scripts. Integrator agents built the project, ran the census of forbidden constructs and the kernel replay.
(5) *Literature.* The literature search of §1.6 was carried out by AI agents.
(6) *This paper* was written by AI agents from the informal proof, the Lean repository and the cited papers. The numerical values in it were copied from the exact computations of the project.

11.2. **What has been verified, and how.**

- *Machine-checked:* the formal statement of §10.1, in terms of the six definitions of §10.2, from the three standard axioms, by the Lean kernel, with a module-by-module replay by `leanchecker`.
- *Cited, not machine-checked:* the identification of the Lean definition `zeta2 7` with the Kubota–Leopoldt value $\zeta_2(7)$ (Lemma 2.3, from [17]).
- *Checked by AI agents only:* the informal proof as written, in particular the sharper constants that the formal proof does not use ($U^* = 0.70670223$, $E^* = 3.8808\ldots$), the numerical remarks, and the certification programs of Section 9; these programs were written by AI agents and have not been reviewed by a human.
- *Not done:* no human expert has refereed the mathematics, the Lean statements or this paper.

11.3. **Suggestions for a human referee.** Since the formal proof covers the Main Theorem completely, a referee who accepts Lean and Mathlib needs to check (i) that the six definitions of §10.2 define the Volkenborn integrals J_s and the number $J_6/768$; (ii) that the statement of §10.1 expresses the Main Theorem for this number; and (iii) Lemma 2.8 of [17]. For the informal proof, the new mathematics is the denominator bound of Section 5: the class lemma (Lemma 5.3) is a residue identity, and the weight of the argument lies in the continuum reduction (Lemmas 5.11–5.12) and in the exact computation of f on 1449 intervals. Theorems V, G and N are standard in method, and Theorem N reduces to the constant $\theta_0 = -\frac{81}{224}$.

Acknowledgements. The formal proof relies on Mathlib, and on the Lean formalization of the Wiener–Ikehara theorem by T. Tao and the contributors of the PrimeNumberTheoremAnd project [26].

References

Theorem and lemma numbers refer to arXiv:2304.00816v1 for [14], arXiv:2306.10393v2 for [16], and arXiv:2505.05005 (v1 and v2 have the same numbering) for [17].

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